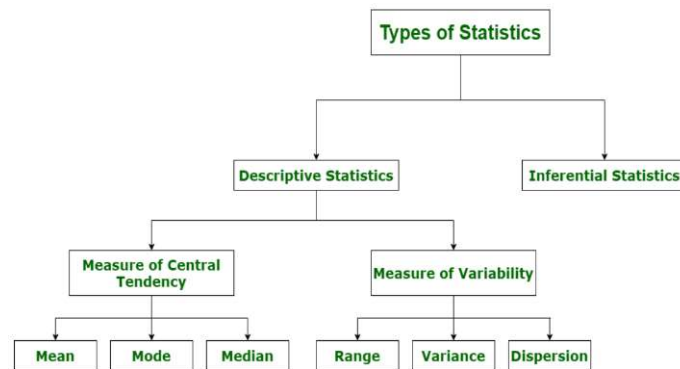


What is Statistics?

- Statistics is a branch of applied mathematics that deals with collection, organisation, analysis of data, and interpretation of results obtained.
- According to Fisher (1947), the science of statistics is essentially a branch of applied mathematics and may be regarded as mathematics, applied to observational data.
- According to Croxton and Cowden, Statistics is the science, which deals with the collection, analysis and interpretation of numerical data.



Descriptive Statistics:

Descriptive statistics consists of methods for organising, displaying, and describing data by using tables, graphs, and summary measures.

Example: A bowler wants to find his bowling average for the past 12 months.

Inferential Statistics:

Inferential statistics consists of methods that use sample results to help make decisions or predictions about a population.

Example: A bowler wants to estimate his chance of winning a game based on his current season averages and the average of his opponents.

Population and Sample

Population:

An aggregate of all individuals or objects defined on some common characteristics is called a population.

- Finite Population
 - Existent
 - Non existent
- Infinite Population
 - Existent
 - Non existent

Sample:

A representative small part of a population is called a sample.

Parameter:

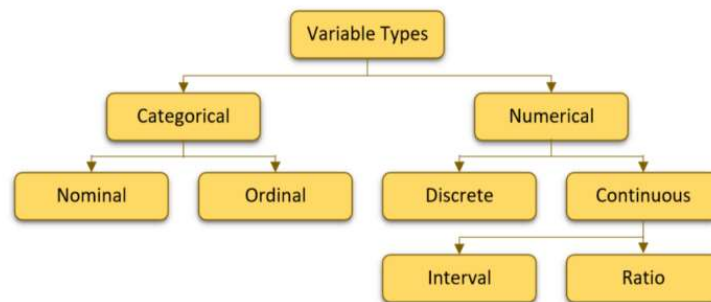
Any population characteristic is called a parameter.

Statistic:

Any sample characteristic is called a statistic.

Variable:

A variable is a measurable quantity, which can vary within its domain.

**Data:**

Data are the raw, disorganised facts and figures collected from any field of inquiry.

- Primary data
- Secondary data

Measurement:

Measurement is a process of assigning numbers to variables or events according to scientific rules. Example: *Family size*.

Scales of measurement is how variables are defined and categorised.

There are 4 types of measurement scale

- *Nominal Scale*
- *Ordinal Scale*
- *Interval Scale*
- *Ratio Scale*

Nominal scale:

The measurement scale, which is used only for identification.

Example: *Eye colour, Gender, Religion, Region etc.*

Ordinal scale:

The measurement scale, which is used for identification as well as ordering.

Example: *Economic Status, Education level, Income level etc.*

Interval scale:

The measurement scale, which is exact but not absolute, is called interval scale.

Example: *Temperature, Test score etc.*

Ratio scale:

The measurement scale, which is exact and absolute is called ratio scale

Example: *Family size, height, weight etc.*

Data Representation:

Data presentation refers to the process of organising, summarising, and visually representing data in a way that is easy to understand and interpret.

Graphical presentation of Qualitative data

✓ Bar diagram.

✓ Pie chart.

Bar diagram:

A bar diagram is a chart or graph that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally.

A bar graph shows comparisons among discrete categories. One axis of the chart shows the specific categories being compared, and the other axis represents a measured value. Some bar graphs present bars clustered in groups of more than one, showing the values of more than one measured variable.

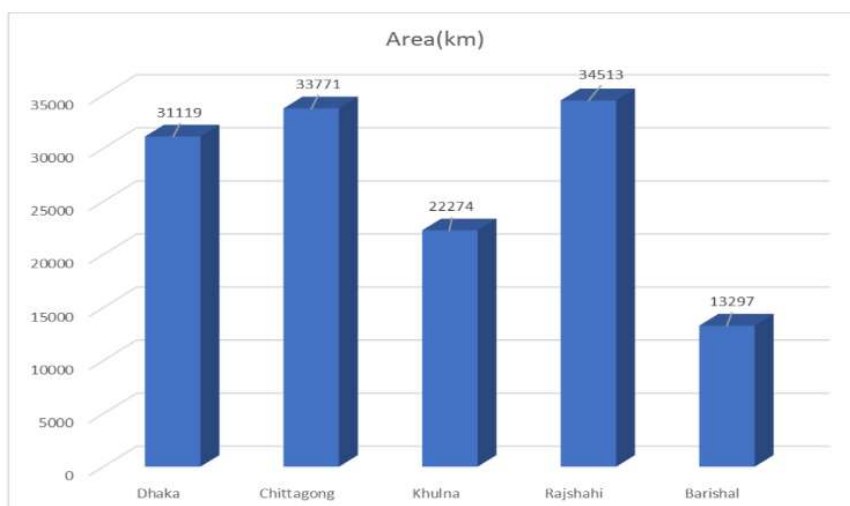
Region-wise area of Bangladesh (in sq.km)

Region	Area(km)
Dhaka	31119
Chittagong	33771
Khulna	22274
Rajshahi	34513
Barisal	13297

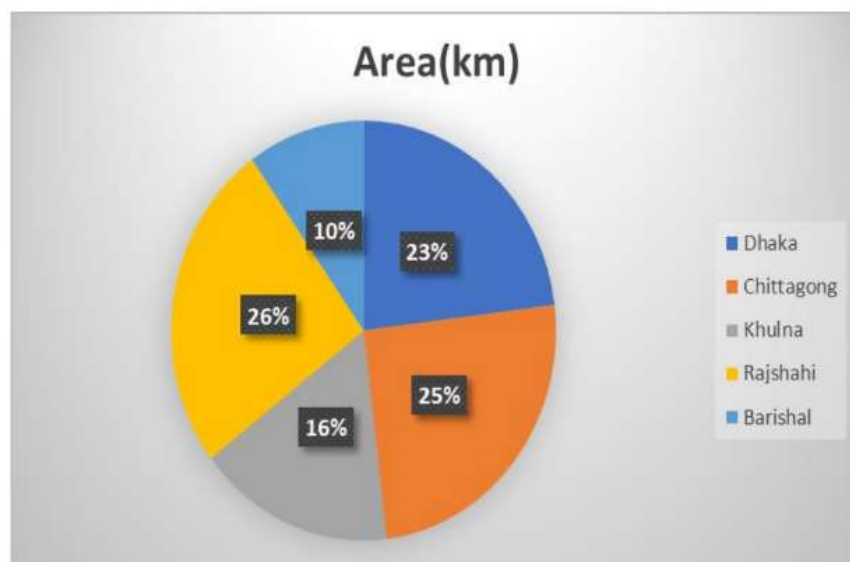
Source: Statistical year Book of Bangladesh, 1996, BBS

Simple Bar Diagram:

Simple bar graphs are a graphical representation of a data set based on one variable. It is widely used to compare various items/ observations/ categories based on a particular parameter. Bar graphs make it easier to compare things as you can analyse and interpret the data just by a glance.

**Pie Chart:**

A pie chart is a type of graph that represents the data in the circular graph. The slices of pie show the relative size of the data, and it is a type of pictorial representation of data.



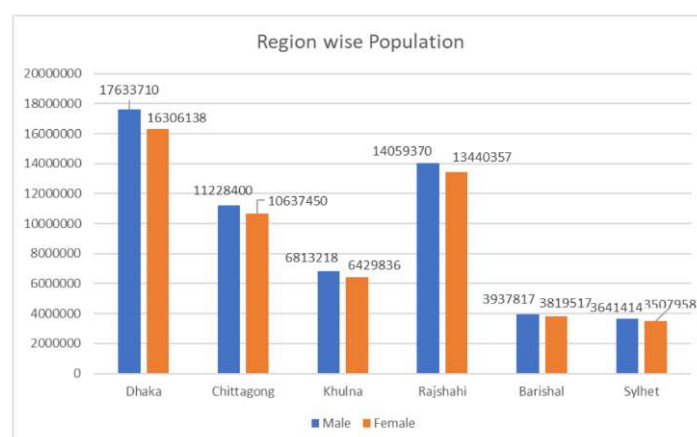
Region-wise Population of Bangladesh by Sex (Census, 1991):

Region	Male	Female
Dhaka	17633710	16306138
Chittagong	11228400	10637450
Khulna	6813218	6429836
Rajshahi	14059370	13440357
Barisal	3937817	3819517
Sylhet	3641414	3507958

Source: Statistical year Book of Bangladesh, 1996, BBS

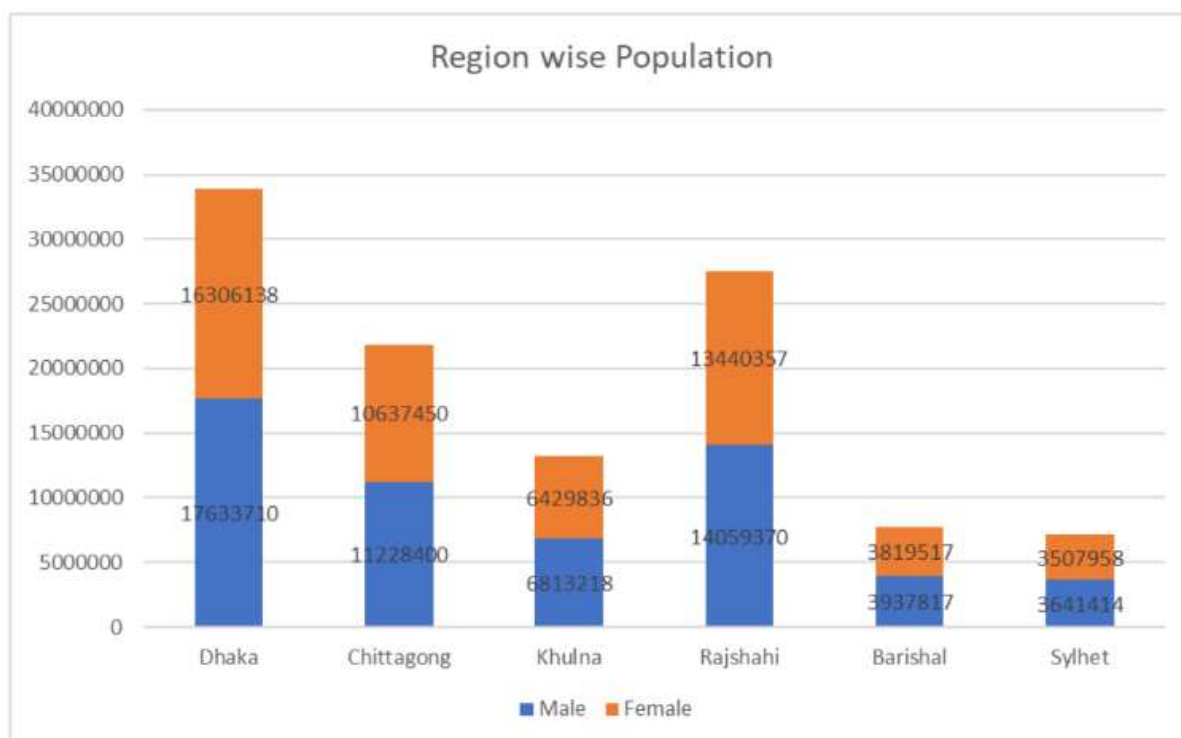
Multiple Bar Diagram:

Multiple bar diagrams are identical to a regular bar graph with the exception that there are two or more bars in each category, one for each subdivision.



Component Bar Diagram:

This type of bar diagram shows the total frequency as well as the various components constituting the total value by a single bar.



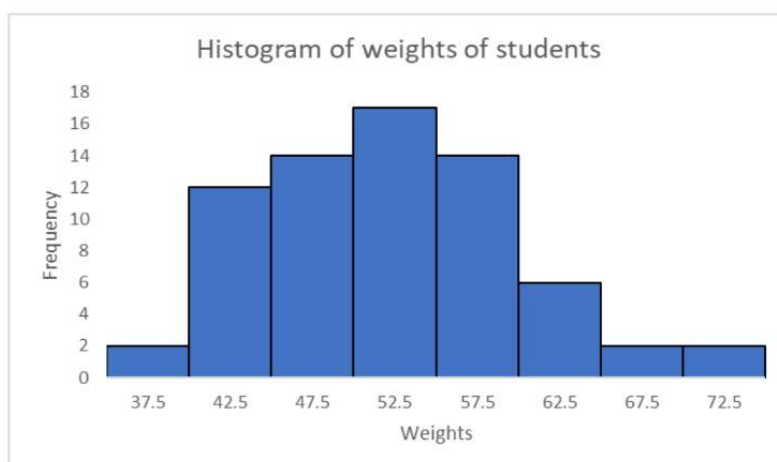
Graphical presentation of Quantitative data

- ✓ Histogram
- ✓ Frequency polygon
- ✓ Cumulative frequency curve or Ogive

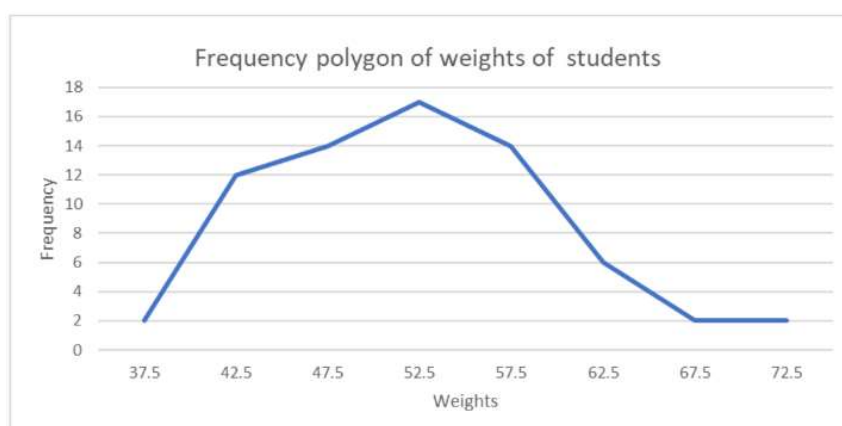
Class boundary	Mid value	frequency	cumulative frequency
35-40	37.5	2	2
40-45	42.5	12	14
45-50	47.5	14	28
50-55	52.5	17	45
55-60	57.5	14	59
60-65	62.5	6	65
65-70	67.5	2	67
70-75	72.5	2	69
Total		69	

Histogram:

In statistics, a histogram is a graphical representation of the distribution of data. The histogram is represented by a set of rectangles, adjacent to each other, where each bar represents a kind of data. A Frequency distribution can be shown graphically by using different types of graphs and a Histogram is one among them.

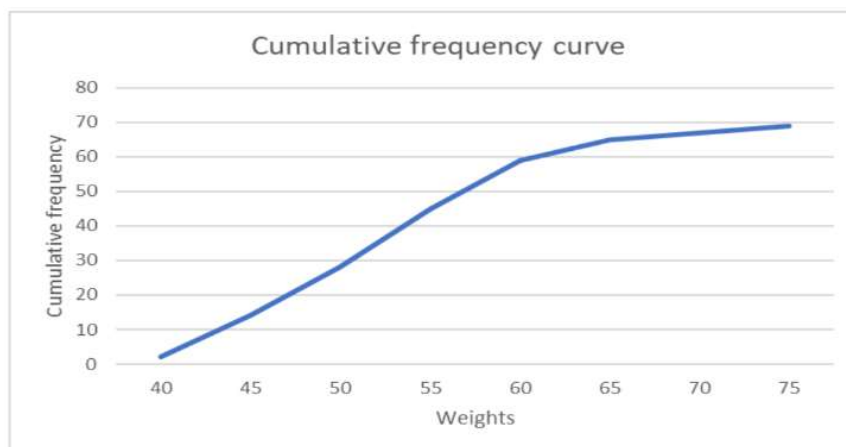
**Frequency polygon:**

A frequency polygon is a line graph of class frequency plotted against class midpoint. It can be obtained by joining the midpoints of the tops of the rectangles in the histogram.



Cumulative frequency curve or Ogive:

The cumulative frequency polygon is essentially a line graph drawn on graph paper by plotting actual lower or upper limits of the class intervals on the x-axis and the respective cumulative frequencies of these class intervals on the y-axis.



Contingency table:

A contingency table displays frequencies for combinations of two categorical variables. Analysts also refer to contingency tables as crosstabulation and two-way tables. Contingency tables classify outcomes for one variable in rows and the other in columns. The values at the row and column intersections are frequencies for each unique combination of the two variables.

The contingency table example below displays computer sales at our fictional store. Specifically, it describes sales frequencies by the customer's gender and the type of computer purchased. It is a two-way table (2×2).

#

Gender	PC	Mac	Total
Male	66	40	106
Female	30	87	117
Total	96	127	223

- (i) Proportion of male among total respondents?
- (ii) Proportion of females among total respondents?
- (iii) Proportion of respondents who purchased a PC?
- (iv) Proportion of respondent who purchased Mac?
- (v) Proportion of male respondent who purchased PC?
- (vi) Proportion of respondents who purchased PCs among females?

Measures of Central Tendency

A measure of **central tendency** is a single value that attempts to describe a set of data by identifying the central position within that set of data. As such, measures of central tendency are sometimes called *measures of central location*. They are also classed as *Summary statistics*. The mean (often called the average) is most likely the measure of central tendency that you are most familiar with, but there are others, such as the median and the mode.

Arithmetic Mean (AM):

Mean is the most commonly used measure of central tendency. It actually represents the average of the given collection of data. It is applicable for both continuous and discrete data.

[For ungrouped data]

Suppose $x_1, x_2, x_3, \dots, x_n$ is n values of a variable X , then the Arithmetic Mean, denoted by $\bar{x}$, is defined as:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

$$= \frac{\sum_1^n x_i}{n}$$

[For grouped data]

If $x_1, x_2, x_3, \dots, x_n$ are the mid-values of n classes and $f_1, f_2, f_3, \dots, f_n$ are the corresponding frequencies such that:

$$f_1 + f_2 + f_3 + \dots + f_n = n \quad \text{Then,}$$

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + f_3 x_3 + \dots + f_n x_n}{f_1 + f_2 + f_3 + \dots + f_n}$$

$$= \frac{\sum_1^n f_i x_i}{\sum_1^n f_i}$$

Median (Me):

Generally median represents the mid-value of the given set of data when arranged in a particular order.

[For ungrouped data]

Step 1. Arrange the given values in ascending order.

Step 2. Find the number of observations in the given set of data. It is denoted by n .

Step 3. If n is **odd**, the median equals the $[(n+1)/2]^{\text{th}}$ observation.

Step 4. If n is **even**, then the median is given by the mean of $[n/2]^{\text{th}}$ observation and $[(n/2)+1]^{\text{th}}$ observation.

[For grouped data]

$$Me = l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$$

Here,

l = lower limit of the median class.

n = no. of observations.

cf = Cumulative frequency of the class preceding the median class.

f = frequency of median class.

h = class size

Mode (Mo):

The most frequent number occurring in the data set is known as the mode.

[For ungrouped data]

To find the mode for ungrouped data, it would be better to arrange the data values either in *ascending* or *descending* order, so that we can easily find the repeated values and their frequency. Hence, the observation with the highest frequency will be the mode of the given data.

[For grouped data]

in a grouped frequency distribution, unlike ungrouped data, it is impossible to determine the mode by looking at the frequencies. Here, we can only locate a class with the maximum frequency, called the **Modal class**. The mode is a value that lies in the modal class and is calculated using the formula given as:

$$Mo = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

Here,

l = Lower limit of the modal class

h = Size of the modal class interval

f_1 = Frequency of the modal class

f_0 = Frequency of the class preceding the modal class

f_2 = Frequency of the class succeeding the modal class

Example:Q-1

Find the arithmetic mean of 4, 8, 12, 16, 20.

Solution:

$$\text{Sum of values } \sum_1^n x_i = 4 + 8 + 12 + 16 + 20 = 60$$

$$\text{Number of values } n = 5$$

$$\bar{x} = \frac{\sum_1^n x_i}{n} = \frac{60}{5} = 12$$

Example:Q-2

Find the median for the given data. The heights (in cm) of 11 players of a team are as follows: 173, 158, 158, 159, 160, 160, 165, 166, 171, 170, 163.

Solution:

On arranging the observations in ascending order, we get

158, 158, 159, 160, 160, 163, 165, 166, 170, 171, 173

There are 11 observations, and the number is odd.

Therefore,

$$\text{Me} = \left[\frac{11+1}{2} \right]^{th} \text{ observation} = 6^{th} \text{ observation} = 163.$$

Example:Q-3

Find the median of the first ten even numbers.

Solution:

First ten odd integers = 2,4,6,8,10,12,14,16,18,20

There are 10 observations, and the number is even

Therefore,

$$\text{Me} = \text{mean of } \left[\frac{10}{2} \right]^{th} = 5^{th} \text{ observation and } \left[\frac{10}{2} + 1 \right]^{th} = 6^{th} \text{ observation.}$$

That means,

$$\text{Me} = \frac{10+12}{2} = 11$$

Example:Q-4

Find the mode of the following marks obtained by 25 students in a mathematics test out of 50.

34, 46, 45, 39, 43, 22, 27, 37, 46, 35, 34, 39, 40, 30, 30, 41, 37, 46, 39, 29, 34, 39, 35, 43, 30

Solution:

The ascending order of the data:

22, 27, 29, 30, 30, 30, 34, 34, 34, 35, 35, 37, 37, 39, 39, 39, 39, 40, 41, 43, 43, 45, 46, 46, 46

The most frequently occurred value is 39.

Hence, the mode of given marks is 39.

Example:Q-5

Find the mean, median and mode for the given data.

Class	Mid value (x_i)	Freq (f_i)	$f_i x_i$	Cumulative freq
2.735-3.135	2.935	6	17.61	6
3.135-3.535	3.335	7	23.345	13
3.535-3.935	3.735	12	44.82	25
3.935-4.335	4.135	12	49.62	37
4.335-4.735	4.535	10	45.35	47
4.735-5.135	4.935	7	34.545	54
5.135-5.535	5.335	3	16.005	57
Total		$\sum_1^n f_i = 57$	$\sum_1^n f_i x_i = 231.295$	

Solution:

$$\begin{aligned}
 \text{AM} &= \frac{\sum_1^n f_i x_i}{\sum_1^n f_i} & \text{Me} &= l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h \\
 &= \frac{231.295}{57} & &= 3.935 + \left(\frac{\frac{57}{2} - 25}{12} \right) \times .4 \\
 &= 4.0578 & &= 4.052
 \end{aligned}$$

From the data we can see that, there exist two modal class

By considering 3.935-4.335 as modal class

$$\begin{aligned}
 \text{Mo} &= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \\
 &= 3.935 + \left(\frac{12 - 12}{2 \times 12 - 12 - 10} \right) \times .4 \\
 &= 3.935
 \end{aligned}$$

By considering 3.535-3.935 as modal class

$$\begin{aligned}
 \text{Mo} &= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \\
 &= 3.535 + \left(\frac{12 - 7}{2 \times 12 - 7 - 12} \right) \times .4 \\
 &= 3.935
 \end{aligned}$$

The mode is same.

So, there is no problem if there exist two modal class.

Example:Q-6

Find the mode for the given data

Class	Freq (f_i)	Cumulative freq
101.005-120.005	2	2
120.005-145.005	6	8
145.005-148.005	12	20
148.005-150.005	23	43
150.005-152.605	11	54
152.605-154.005	9	63
154.005-155.005	4	67
155.005-156.005	2	69
Total	69	

$$\begin{aligned}
 Mo &= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \\
 &= 148.005 + \left(\frac{23 - 12}{2 \times 23 - 12 - 11} \right) \times 2 \\
 &= 148.962
 \end{aligned}$$

Other Average**Geometric Mean (GM):**

In Mathematics, the *Geometric Mean (GM)* is the average value or mean which signifies the central tendency of the set of numbers by finding the product of their values. Basically, we multiply the numbers altogether and take the n th root of the multiplied numbers, where n is the total number of data values.

Suppose $x_1, x_2, x_3, \dots, \dots, \dots, x_n$ is n values of a variable X , then the GM, is defined as:

$$GM = \sqrt[n]{x_1 \times x_2 \times x_3 \times \dots \times x_n}$$

Or,

$$GM = (x_1 \times x_2 \times x_3 \times \dots \times x_n)^{\frac{1}{n}}$$

$$\text{Log } Gm = \frac{1}{n} \log(x_1 \times x_2 \times x_3 \times \dots \times x_n)$$

$$= \frac{1}{n} (\log x_1 + \log x_2 + \log x_3 + \dots + \log x_n)$$

$$= \frac{\sum_1^n \log x_i}{n}$$

$$GM = \text{Antilog} \frac{\sum_1^n \log x_i}{n}$$

This is the formula of GM for ungrouped data

Example:Q-7

Find the GM of the values 10, 25, 5, and 30

Solution:

We know that,

$$GM = \sqrt[n]{x_1 \times x_2 \times x_3 \times \dots \times x_n}$$

$$= \sqrt[4]{10 \times 25 \times 5 \times 30}$$

$$= 13.92$$

[For grouped data]

$$GM = \text{Antilog} \frac{\sum_1^n f_i \log x_i}{\sum_1^n f_i}$$

Example:Q-8

Find the GM for the given data.

Marks	Mid point x_i	f_i	$\log x_i$	$f_i \log x_i$
4-8	6	6	0.7782	4.6692
8-12	10	10	1.0000	10.0000
12-16	14	18	1.1461	20.6298
16-20	18	30	1.2553	37.6590
20-24	22	15	1.3424	20.1360
24-28	26	12	1.4150	16.800
28-32	30	10	1.4771	14.7710
32-36	34	6	1.5315	9.1890
36-40	38	2	1.5798	3.1596
Total		$\sum_1^n f_i = 109$		$\sum_1^n f_i \log x_i$ $= 137.1936$

Solution:

$$\begin{aligned}
 \text{GM} &= \text{Antilog} \frac{\sum_1^n f_i \log x_i}{\sum_1^n f_i} \\
 &= \text{Antilog} \frac{137.1936}{109} \\
 &= 18.14
 \end{aligned}$$

Harmonic Mean (HM):

Suppose $x_1, x_2, x_3, \dots, x_n$ is n values of a variable X , then the HM, is defined as:

[For ungrouped data]

$$HM = \frac{n}{\sum_1^n \frac{1}{x_i}}$$

[For grouped data]

$$HM = \frac{\sum_1^n f_i}{\sum_1^n \frac{f_i}{x_i}}$$

Weighted Mean:

Weighted Mean is an average computed by giving different weights to some of the individual values.

Suppose $x_1, x_2, x_3, \dots, x_n$ is n values of a variable X and each of these values is weighted by the amount $W_1, W_2, W_3, \dots, W_n$ ($w_i > 0, \forall i = 1, 2, \dots, n$), respectively.

Weighted Arithmetic Mean (WAM):

$$WAM = \frac{\sum_1^n w_i x_i}{\sum_1^n w_i}$$

Weighted Geometric Mean (WGM)

$$WGM = \text{Antilog} \frac{\sum_1^n w_i \log x_i}{\sum_1^n w_i}$$

Weighted Harmonic Mean (WHM)

$$WHM = \frac{\sum_1^n w_i}{\sum_1^n \frac{w_i}{x_i}}$$

Quartiles, Deciles and Percentiles:

[For ungrouped data]

Quartiles

$$Q_i = \left(\frac{i \cdot (n+1)}{4} \right)^{th} \text{ value of the observation, where } i=1,2,3$$

Deciles

$$D_i = \left(\frac{i \cdot (n+1)}{10} \right)^{th} \text{ value of the observation, where } i=1,2,3, \dots, 9$$

Percentiles

$$P_i = \left(\frac{i \cdot (n+1)}{100} \right)^{th} \text{ value of the observation, where } i=1,2,3, \dots, 99$$

[For grouped data]

Quartiles

$Q_i \text{ class} = \left(\frac{i \cdot n}{4}\right)^{th}$ value of the observation

$Q_i = l + \left(\frac{\frac{i \cdot n}{4} - cf}{f}\right) \times h$, where $i=1,2,3$

Deciles

$D_i \text{ class} = \left(\frac{i \cdot n}{10}\right)^{th}$ value of the observation

$D_i = l + \left(\frac{\frac{i \cdot n}{10} - cf}{f}\right) \times h$, where $i=1,2,3, \dots, 9$

Percentiles

$P_i \text{ class} = \left(\frac{i \cdot n}{100}\right)^{th}$ value of the observation

$P_i = l + \left(\frac{\frac{i \cdot n}{100} - cf}{f}\right) \times h$, where $i=1,2,3, \dots, 99$

Example:09

Calculate Quartile-1, Deciles-3, Percentiles-20 from the following data
3,13,11,11,5,4,2.

Solution:

Arranging Observations in the ascending order, we get :

2,3,4,5,11,11,13

Here, $n=7$

$Q_1 = \left(\frac{1 \cdot (7+1)}{4}\right)^{th}$ value of the observation

$= 2^{nd}$ value of the observation

$= 3$

$D_3 = \left(\frac{3 \cdot (7+1)}{10}\right)^{th}$ value of the observation

$= 2.4^{th}$ value of the observation

$= 2^{nd} \text{ observation} + 0.4[3^{rd} - 2^{nd}]$

$= 3 + 0.4[4 - 3]$

$= 3 + 0.4(1)$

$= 3 + 0.4$

$= 3.4$

$$\begin{aligned}
P_{20} &= \left(\frac{20 \cdot (7+1)}{100} \right)^{th} \text{ value of the observation} \\
&= 1.6^{th} \text{ value of the observation} \\
&= 1^{\text{st}} \text{ observation} + 0.6[2^{\text{nd}} - 1^{\text{st}}] \\
&= 2 + 0.6[3 - 2] \\
&= 2 + 0.6(1) \\
&= 2 + 0.6 \\
&= 2.6
\end{aligned}$$

Measures of Dispersion:

In statistics, the measures of dispersion help to interpret the variability of data i.e. to know how *homogeneous* or *heterogeneous* the data is. In simple terms, it shows how scattered the variable is.

Types of Measures of Dispersion:

There are two main types of dispersion methods in statistics which are:

- ✓ Absolute Measure of Dispersion
- ✓ Relative Measure of Dispersion

Absolute Measure of Dispersion:

An absolute measure of dispersion contains the same unit as the original data set. The absolute dispersion method expresses the variations in terms of the average of deviations of observations like standard or mean deviations. It includes range, standard deviation, quartile deviation, etc.

The types of absolute measures of dispersion are:

- ✓ Range
- ✓ Variance
- ✓ Standard Deviation
- ✓ Quartile Deviation
- ✓ Mean Deviation

Range:

It is simply the difference between the maximum value and the minimum value given in a data set.

Example: 1, 3, 5, 6, 7

Here, Range = 7 - 1 = 6

Variance:

The term variance refers to a statistical measurement of the spread between numbers in a data set. More specifically, variance measures how far each number in the set is from the mean (average), and thus from every other number in the set. Variance is often depicted by this symbol: σ^2

In our study, we have two types of variances,

- Population variance: Let, $x_1, \dots, x_N$ be N observations in a population and μ be the population mean. Then the population variance denoted by σ^2 is defined as,

$$\sigma^2 = \sum_1^N \frac{(x_i - \mu)^2}{N}$$

For grouped data
$$\sigma^2 = \sum_1^N \frac{f_i(x_i - \mu)^2}{\sum_1^n f_i}$$

- Sample variance: Let, $x_1, \dots, x_n$ be n observations in a sample and $\bar{x}$ be the sample mean. Then the sample variance denoted by s^2 is defined as,

$$s^2 = \sum_1^n \frac{(x_i - \bar{x})^2}{n - 1}$$

For grouped data
$$s^2 = \sum_1^n \frac{f_i(x_i - \bar{x})^2}{\sum_1^n f_i - 1}$$

Standard deviation:

The positive square root of variance is called standard deviation.

- Population standard deviation $\sigma = \sqrt{\sum_1^N \frac{(x_i - \mu)^2}{N}}$

For grouped data
$$\sigma = \sqrt{\sum_1^N \frac{f_i(x_i - \mu)^2}{\sum_1^n f_i}}$$

- Sample standard deviation $s = \sqrt{\sum_1^n \frac{(x_i - \bar{x})^2}{n - 1}}$

For grouped data
$$s = \sqrt{\sum_1^n \frac{f_i(x_i - \bar{x})^2}{\sum_1^n f_i - 1}}$$

Example:10

Find the variance and standard deviation for the following data.

Class	Mid value (x_i)	Freq (f_i)	$f_i x_i$	$(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^2$
2.735-3.135	2.935	6	17.61	1.26068	7.564079
3.135-3.535	3.335	7	23.345	0.52244	3.657079
3.535-3.935	3.735	12	44.82	0.1042	1.250398
3.935-4.335	4.135	12	49.62	0.00596	0.071518
4.335-4.735	4.535	10	45.35	0.22772	2.277198
4.735-5.135	4.935	7	34.545	0.76948	5.386359
5.135-5.535	5.335	3	16.005	1.63124	4.89372
Total		$\sum_1^n f_i = 57$	$\sum_1^n f_i x_i = 231.295$	4.521719	$\sum_1^n f_i(x_i - \bar{x})^2 = 25.10035$

$$\text{Sample mean } \bar{x} = \frac{\sum_1^n f_i x_i}{\sum_1^n f_i}$$

$$= \frac{231.295}{57}$$

$$= 4.0578$$

$$\begin{aligned} \text{Sample variance } s^2 &= \frac{\sum_1^n f_i(x_i - \bar{x})^2}{\sum_1^n f_i - 1} \\ &= \frac{25.10035}{57-1} \\ &= 0.448221 \end{aligned}$$

$$\begin{aligned} \text{Sample standard deviation } s &= \sqrt{0.448221} \\ &= 0.669 \end{aligned}$$

Mean Deviation:

Mean deviation is used to compute how far the values in a data set are from the center point. Mean, median, and mode all form center points of the data set. In other words, the mean deviation is used to calculate the average of the absolute deviations of the data from the central point.

In case of mean:

$$\text{M.A.D } (\bar{x}) = \frac{\sum_1^n |x_i - \bar{x}|}{n}$$

$$\text{For grouped data } \text{M.A.D } (\bar{x}) = \frac{\sum_1^n f_i |x_i - \bar{x}|}{\sum_1^n f_i}$$

In case of median:

$$\text{M.A.D } (Me) = \frac{\sum_1^n |x_i - Me|}{n}$$

$$\text{For grouped data } \text{M.A.D } (Me) = \frac{\sum_1^n f_i |x_i - Me|}{\sum_1^n f_i}$$

In case of mode:

$$\text{M.A.D } (Mo) = \frac{\sum_1^n |x_i - Mo|}{n}$$

For grouped data
$$\text{M.A.D } (Mo) = \frac{\sum_1^n f_i |x_i - Mo|}{\sum_1^n f_i}$$

Example:11

Find the mean deviation for the following data.

Class	Mid value (x_i)	Freq (f_i)	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
15-25	20	25	500	13.68	342
25-35	30	54	1620	3.68	198.72
35-45	40	34	1360	6.32	214.88
45-55	50	20	1000	16.32	326.4
Total		$\sum_1^n f_i = 133$	$\sum_1^n f_i x_i = 4480$		$\sum_1^n f_i x_i - \bar{x} = 1082$

$$\text{Sample mean } \bar{x} = \frac{\sum_1^n f_i x_i}{\sum_1^n f_i}$$

$$= \frac{4480}{133}$$

$$= 33.68$$

$$\text{M.A.D } (\bar{x}) = \frac{\sum_1^n f_i |x_i - \bar{x}|}{\sum_1^n f_i}$$

$$= \frac{1082}{133}$$

$$= 8.14$$

Relative Measure of Dispersion:

The relative measures of dispersion are used to compare the distribution of two or more data sets. This measure compares values without units.

Common relative dispersion methods include:

- ✓ Coefficient of Range
- ✓ Coefficient of Variation (C.V)
- ✓ Coefficient of Standard Deviation
- ✓ Coefficient of Quartile Deviation
- ✓ Coefficient of Mean Deviation

Coefficient of variation:

The coefficient of variation is the ratio of the standard deviation to the mean. It is usually expressed as percentage. Mathematically,

$$CV = \frac{\sigma}{\mu} \times 100; \text{For population}$$

$$CV = \frac{s}{\bar{x}} \times 100; \text{For Sample}$$

Example:12

Find the coefficient of variation for the two plants of a factory for the given data and interpret the results.

Two plants C and D of a factory show the following results about the number of workers and the wages paid to them.

No. of workers	5000	6000
Average monthly wages	\$2,500	\$2,500
Standard deviation	9	10

Solution:

To Find: Which plant has greater variability.

For this, we need to find the coefficient of variation. The plant that has a higher coefficient of variation will have greater variability. Using coefficient of variation formula,

$$CV = \frac{s}{\bar{x}} \times 100$$

Coefficient of variation for plant C

$$CV = (9/2500) \times 100$$

$$CV = 0.36\%$$

Now, Coefficient of variation for plant D

$$CV = (10/2500) \times 100$$

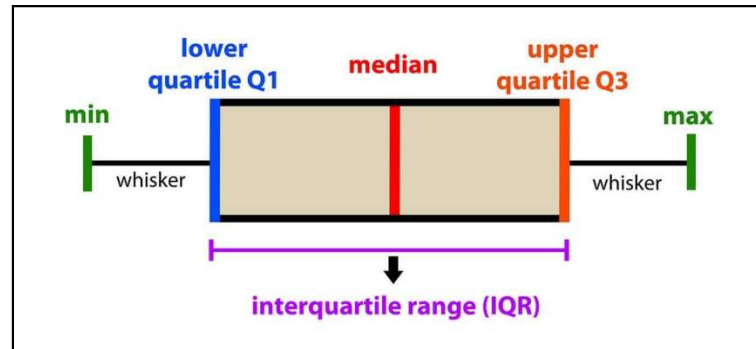
$$CV = 0.4\%$$

Plant C has CV = 0.36 and plant D has CV = 0.4

Hence plant D has greater variability in individual wages.

Box plot:

When we display the data distribution in a standardised way using 5 summary – minimum, Q1 (First Quartile), Q2 (median) Q3 (third Quartile), and maximum, it is called a Box plot. It is also termed as box and whisker plot.

**Minimum Score:**

The lowest score, excluding outliers (shown at the end of the left whisker).

Lower Quartile:

Twenty-five percent of scores fall below the lower quartile value (also known as the first quartile).

Median:

The median marks the midpoint of the data and is shown by the line that divides the box into two parts (sometimes known as the second quartile). Half the scores are greater than or equal to this value, and half are less.

Upper Quartile:

Seventy-five percent of the scores fall below the upper quartile value (also known as the third quartile). Thus, 25% of data are above this value.

Maximum Score:

The highest score, excluding outliers (shown at the end of the right whisker).

Whiskers:

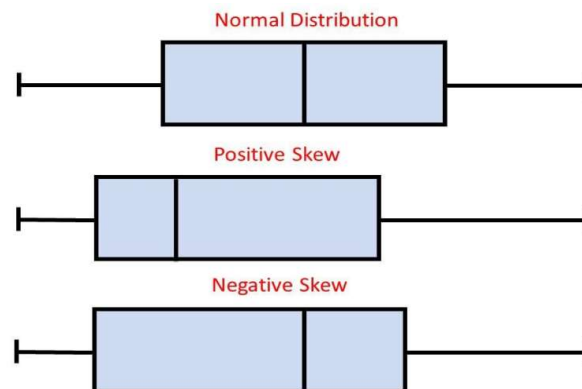
The upper and lower whiskers represent scores outside the middle 50% (i.e., the lower 25% of scores and the upper 25% of scores).

The Interquartile Range (IQR):

The box plot shows the middle 50% of scores (i.e., the range between the 25th and 75th percentile).

Why Are Box Plots Useful?:

- *Box plots are useful as they show the average score of a data set:*
The median is the average value from a set of data and is shown by the line that divides the box into two parts. Half the scores are greater than or equal to this value, and half are less.
- *Box plots are useful as they show the skewness of a data set:*
The box plot shape will show if a statistical data set is normally distributed or skewed.

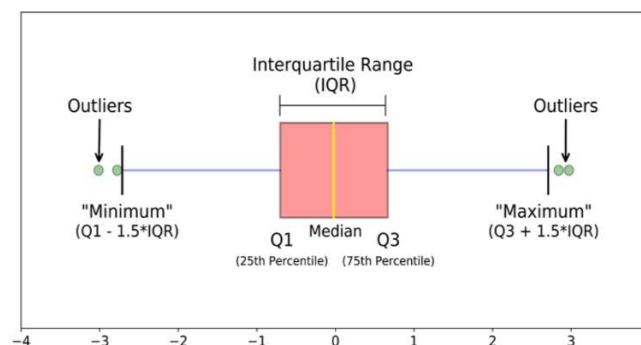


When the median is in the middle of the box, and the whiskers are about the same on both sides of the box, then the distribution is symmetric.

When the median is closer to the bottom of the box, and if the whisker is shorter on the lower end of the box, then the distribution is positively skewed (skewed right).

When the median is closer to the top of the box, and if the whisker is shorter on the upper end of the box, then the distribution is negatively skewed (skewed left).

- *Box plots are useful as they show outliers within a data set:*
An outlier is an observation that is numerically distant from the rest of the data. When reviewing a box plot, an outlier is defined as a data point that is located outside the whiskers of the box plot.



Stem and leaf plot:

A stem and leaf plot, also known as a stem and leaf diagram, is a way to arrange and represent data so that it is simple to see how frequently various data values occur. It is a plot that displays ordered numerical data.

A stem and leaf plot is shown as a special table where the digits of a data value are divided into a stem (first few digits) and a leaf (usually the last digit). The symbol '|' is used to split and illustrate the stem and leaf values. For instance, 105 is written as 10 on the stem and 5 on the leaf.

This can be written as $10 | 5$. Here, $10 | 5 = 105$ is called the key. The key depicts the data value a stem and leaf represent.

Problem 1:

Construct a stem-and-leaf plot for the data in the table.

Cloth Lengths (centimeters)									
5	1	20	12	27	2	30	5	7	38
40	47	1	2	1	32	4	44	33	23

Solution:

Step 1: Sort the data values: 1, 1, 1, 2, 2, 4, 5, 5, 7, 12, 20, 23, 27, 30, 32, 33, 38, 40, 44, 47

Step 2: Choose the stems and the leaves. As the data values range from 1 to 47, use the tens digits for the stems and the ones digits for the leaves. Be sure to include the key.

Step 3: Write the stems to the left of the vertical line from the top to bottom.

Step 4: Write the leaf values corresponding to each stem to the right of the vertical line.

Stem	Plot
0	1 1 1 2 2 4 5 5 7
1	2
2	0 3 7
3	0 2 3 8
4	0 4 7

Key: $0 | 1 = 1 \text{ cm}$

Problem 2:

The stem-and-leaf plot below shows the quiz scores of students.

(a) Find the number of students who scored less than 9 points?

(b) Find the number of students who scored a minimum of 9 points?

Quiz Scores	
Stem	Leaf
6	6
7	0 5 7 8
8	1 1 3 4 4 6 8 8 9
9	0 2 9
10	0

Key: $9 | 2 = 9.2 \text{ points}$

Decimal or fractional values can also be placed in a stem-and-leaf plot.

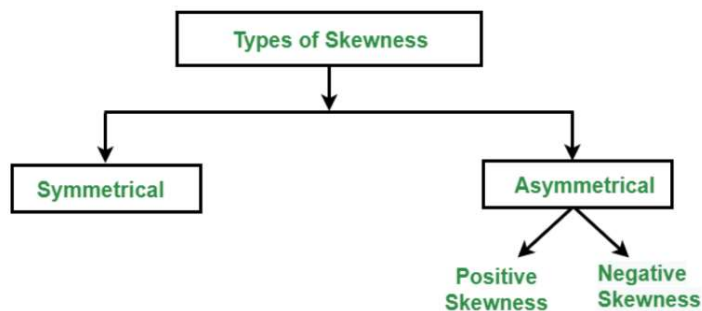
Decimal Between Stem and Leaf	Decimal in the Stem
12.3, 12.5, 13.0	1.23, 1.25, 1.30
Becomes	Becomes
$\begin{array}{c c} 12 & 3, 5 \\ 13 & 0 \end{array}$	$\begin{array}{c c} 1.2 & 3, 5 \\ 1.3 & 0 \end{array}$
Key: 12 3 = 12.3 units	Key: 1.2 3 = 1.23 units

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Skewness and Kurtosis:

Skewness is a measure of the asymmetry of a distribution. Skewness is an important statistical technique that helps to determine asymmetrical behaviour than of the frequency distribution, or more precisely, the lack of symmetry of tails both left and right of the frequency curve. A distribution or dataset is symmetric if it looks the same to the left and right of the centre point.

Types of skewness: The following figure describes the classification of skewness:

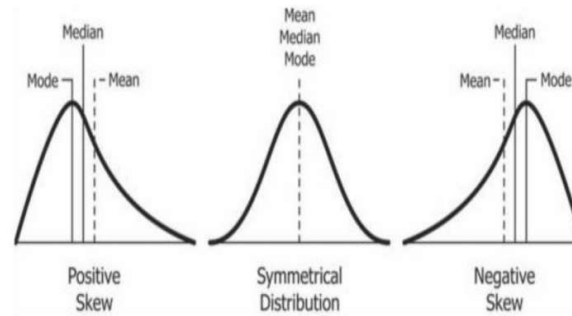


Symmetric Skewness: A perfect symmetric distribution is one in which frequency distribution is the same on the sides of the centre point of the frequency curve. In this, Mean = Median = Mode. There is no skewness in a perfectly symmetrical distribution.

Asymmetric Skewness: A asymmetrical or skewed distribution is one in which the spread of the frequencies is different on both the sides of the centre point or the frequency curve is more stretched towards one side or value of Mean. Median and Mode fall at different points.

Positive Skewness: In this, the concentration of frequencies is more towards higher values of the variable i.e. the right tail is longer than the left tail.

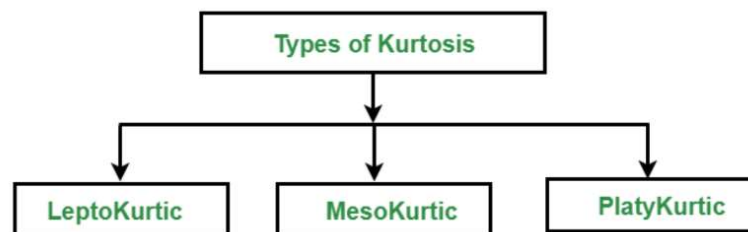
Negative Skewness: In this, the concentration of frequencies is more towards the lower values of the variable i.e. the left tail is longer than the right tail.



Kurtosis:

Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails, or outliers. Data sets with low kurtosis tend to have light tails, or lack of outliers.

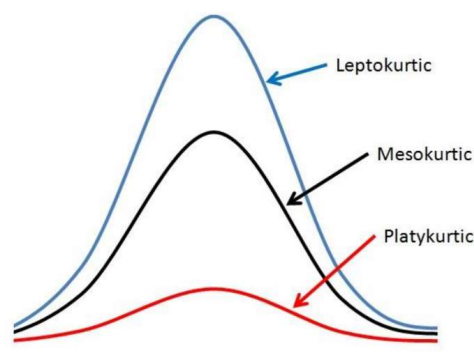
Types of kurtosis: The following figure describes the classification of kurtosis:



Leptokurtic: Leptokurtic is a curve having a higher peak than the normal distribution. In this curve, there is too much concentration of items near the central value.

Mesokurtic: Mesokurtic is a curve having a normal peak than the normal curve. In this curve, there is equal distribution of items around the central value.

Platykurtic: Platykurtic is a curve having a lower peak than the normal curve is called platykurtic. In this curve, there is less concentration of items around the central value.



Correlation:

Correlation Analysis is a statistical method that is used to discover if there is a relationship between two variables and how strong that relationship may be. For example, the height and weight of a person are related, and taller people tend to be heavier than shorter people.

Purpose of correlation analysis:

- i. Degree of relationship.
- ii. Direction of relationship.

Scatter diagram:

The scatter diagram is also called **a scatter plot chart, XY chart, and correlation chart**. A scatter diagram is a two-dimensional graphical representation of a set of data. The scatter diagram graphs pairs numerical data with one variable on each axis to look for a relationship between them. If the variables are correlated, the points will fall along the line or curve. The better the correlation, the tighter the points will hug the line.

Scatter diagrams are used and applied in several ways, where the most important benefit is showing the correlation between two variables. The scatter diagram will visualise in an easy to observe way if the data points are positively correlated, negatively correlated, or there is no correlation between the two variables.

What Are the Types of Scatter Plots?

You can classify a scatter plot in many ways; the most popular one is based on correlation and is extensively used in project management. According to the correlation, scatter plots are divided into the following three categories.

- i. Positive Correlation
- ii. Negative Correlation
- iii. No Correlation

Positive Correlation:

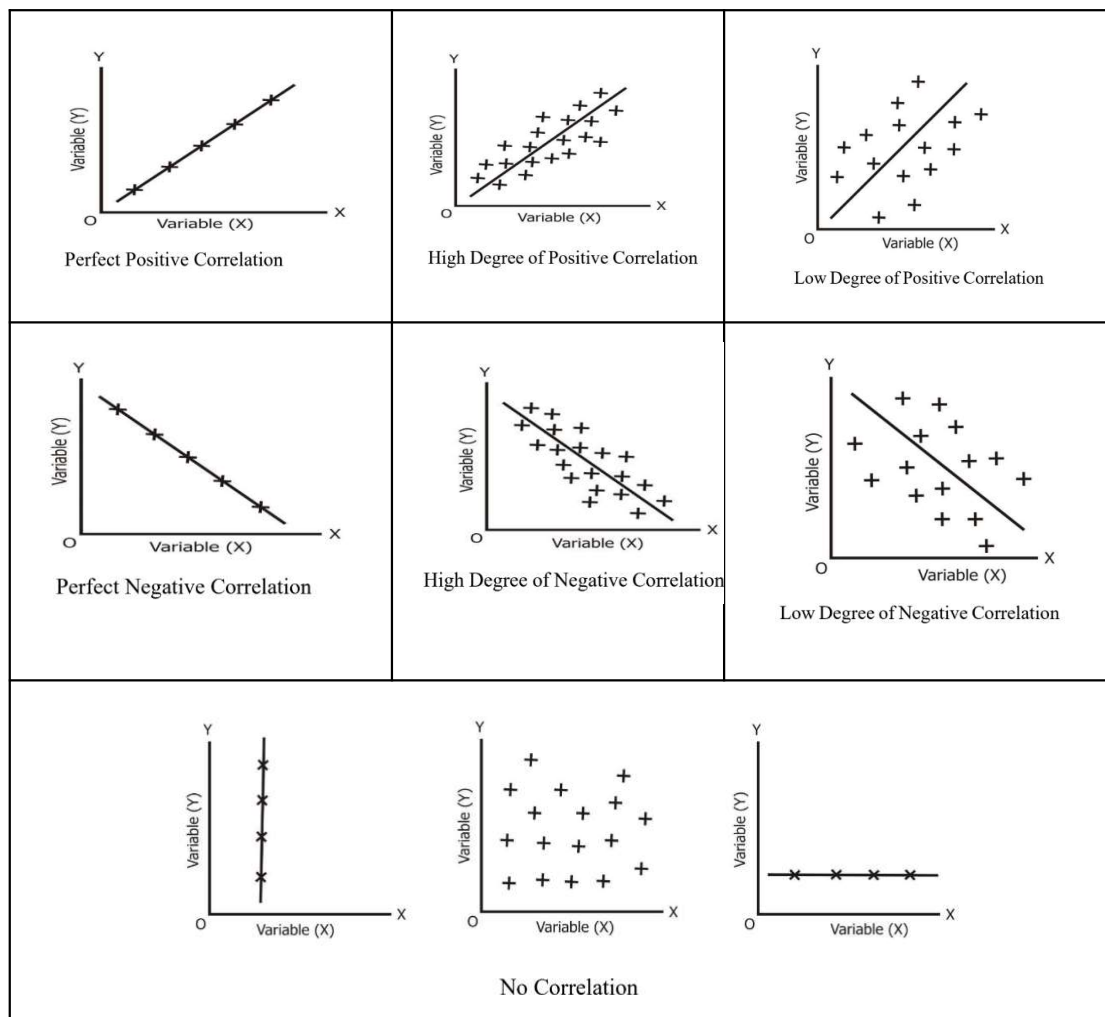
The scatter plot with a positive correlation is also known as a "*Scatter Diagram with Positive Slant*." In this case, as the value of X increases, the value of Y will *increase* too, which means that the correlation between the two variables is positive. If you draw a straight line along the data points, the slope of the line will go up. For example, if the weather gets colder, hot drink sales will go up.

Negative Correlation:

The scatter plot with a negative correlation is also known as a "Scatter Diagram with a Negative Slant." In this case, as the value of X increases, the value of Y will *decrease*. If you draw a straight line along the data points, the slope of the line will go down. For example, if the cycle time of a workflow goes up, the number of tasks completed will go down.

No Correlation:

The scatter plot with no correlation is also known as the "Scatter Diagram with Zero Degree of Correlation." In this case, the data point spreads so randomly that you can't draw a line through the data points. You can conclude that these two variables have no correlation or zero degrees of correlation. For example, if the weather gets hotter, we can't conclude that the sales of wooden chairs will go up or down because there is no correlation between the two variables.



Pearson correlation coefficient:

The Pearson correlation coefficient (r) is the most common way of measuring a linear correlation. It is a number between -1 and 1 that measures the strength and direction of the relationship between two variables.

$$r = \frac{\sum_1^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_1^n (x_i - \bar{x})^2 \sum_1^n (y_i - \bar{y})^2}}$$

Or,

$$r = \frac{\sum_1^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_1^n x_i^2 - n\bar{x}^2)(\sum_1^n y_i^2 - n\bar{y}^2)}}$$

Example:

Math score(X)	70	85	66	15	69	49	73	44	88	69
Overall score(Y)	73	90	50	31	56	55	80	58	85	73

What is the correlation between these two variables?

Solution:

x_i	y_i	$x_i y_i$	x_i^2	y_i^2
70	73	5110	4900	5329
85	90	7650	7225	8100
66	50	3300	4356	2500
15	31	465	225	961
69	56	3864	4761	3136
49	55	2695	2401	3025
73	80	5840	5329	6400
44	58	2552	1936	3364
88	85	7480	7744	7225
69	73	5037	4761	5329
$\sum_1^n x_i = 628$	$\sum_1^n y_i = 651$	$\sum_1^n x_i y_i = 43993$	$\sum_1^n x_i^2 = 43638$	$\sum_1^n y_i^2 = 45369$

$$\begin{aligned} \text{Here, } \hat{x} &= \frac{\sum_1^n x_i}{n} \\ &= \frac{628}{10} \\ &= 62.8 \end{aligned}$$

$$\begin{aligned} \hat{y} &= \frac{\sum_1^n y_i}{n} \\ &= \frac{651}{10} \\ &= 65.1 \end{aligned}$$

$$\begin{aligned} \text{We know, } r &= \frac{\sum_1^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_1^n x_i^2 - n\bar{x}^2)(\sum_1^n y_i^2 - n\bar{y}^2)}} \\ &= \frac{43993 - 10 \cdot 62.8 \cdot 65.1}{\sqrt{(43638 - 10 \cdot 62.8^2)(45369 - 10 \cdot 65.1^2)}} \\ &= 0.878 \end{aligned}$$

Interpretation: The value of 0.878 is close to 1, suggesting a strong positive correlation. This means that there is a strong and positive linear association between maths scores and overall scores.

Regression Analysis:

Regression analysis is a set of statistical methods used for the estimation of relationships between a dependent variable and one or more independent variables. For example,

- ✓ It can be used to predict the relationship between reckless driving and the total number of road accidents.
- ✓ The effect on sales and spending a certain amount of money on advertising.

Purpose of regression analysis:

- i. Cause effect relationship.
- ii. Prediction.

Types of variables in regression analysis:

- Dependent variables.
- Independent variables.

Dependent variables:

The variables where value is influenced or is to be predicted. The dependent variable is also known as *response*, *regress* or *explained variable*.

Independent variables:

The variables which influence the values of the dependent variables or are used for prediction. The independent variable is also known as *explanatory variable*, *predictor*, *covariate* or *regressor*.

Types of regression equation

- i. **Simple regression equation:** A regression equation containing only one independent variable is called a *simple regression equation*.
- ii. **Multiple regression equation:** A regression equation containing more than one independent variable is called *multiple regression equation*.

Simple linear regression equation:

Let Y be the response variable and we want to explain Y by a single explanatory variable X.

The basic model is

$$y = \alpha + \beta x + \varepsilon$$

- $\hat{y}$ is the predicted value of the dependent variable for any given value of the independent variable (x).
- α is the intercept, the predicted value of y when the x is 0.
- β is the regression coefficient (slope) – how much we expect y to change as x increases.
- x is the independent variable (the variable we expect is influencing y).

Here, α and β are parameters that must be estimated. The symbol ε represents the random error term. This does not mean that a mistake is being made. It is simply a symbol used to indicate the absence of an exact relationship between x and y .

Estimated/fitted equation is $\hat{y} = \hat{\alpha} + \hat{\beta}x$

$$\text{Here, } \hat{\beta} = \frac{\sum_1^n x_i y_i - n\bar{x}\bar{y}}{\sum_1^n x_i^2 - n\bar{x}^2}$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Example 1: Exam Scores and Study Hours

Suppose you want to determine if there is a relationship between the number of hours a student studies (independent variable) and their exam scores (dependent variable). You collect data from a sample of students and want to fit a simple linear regression model to the data.

Study hours (X)	2	3	4	5	6
Exam scores (Y)	60	70	75	80	85

Solution:

x_i	y_i	$x_i y_i$	x_i^2
2	60	120	4
3	70	210	9
4	75	300	16
5	80	400	25
6	85	510	36
$\sum_1^n x_i = 20$	$\sum_1^n y_i = 370$	$\sum_1^n x_i y_i = 1540$	$\sum_1^n x_i^2 = 90$

$$\text{Here, } \bar{x} = \frac{\sum_1^n x_i}{n}$$

$$= \frac{20}{5}$$

$$= 4$$

$$\bar{y} = \frac{\sum_1^n y_i}{n}$$

$$= \frac{370}{5}$$

$$= 74$$

$$\text{We know, } \hat{\beta} = \frac{\sum_1^n x_i y_i - n \bar{x} \bar{y}}{\sum_1^n x_i^2 - n \bar{x}^2}$$

$$= \frac{1540 - 5 \cdot 4 \cdot 74}{90 - 5 \cdot 4^2}$$

$$= 6$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

$$= 74 - 6 \cdot 4$$

$$= 50$$

Fitted equation is $\hat{y} = 50 + 6x$

Interpretation:

- The slope ($\beta = 6$) indicates that, on average, for each additional hour of study, the exam score is expected to increase by approximately 6 points.
- The intercept ($\hat{\alpha} = 50$) suggests that if a student doesn't study at all (0 hours), their expected exam score is around 50.

For study hour, $x = 8$ hours (say)

$$\hat{y} = 50 + 6 \cdot 8$$

$$= 98$$

So, the predicted exam score is 98.

Coefficient of Determination:

The coefficient of determination is the square of the Pearson correlation coefficient (r). The coefficient of determination, r^2 , is the proportion of variation in the observed values of the dependent variable explained by the independent variable. The coefficient of determination, r^2 , always lies between 0 and 1. A value of r^2 near 0 suggests that the regression equation is not very useful for making predictions, whereas a value of r^2 near 1 suggests that the regression equation is quite useful for making predictions.

For the above example, $r = 0.9480542$ (say) and $r^2 = 0.899$, that means 89.9% variation in the exam score can be explained by the study hours.

Multiple regression equation:

Suppose, we have k independent variables $x_1, x_2, \dots, x_k$ and we want to explain Y . The basic model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \varepsilon$$

Where, β_0 is the intercept of the regression equation.

β_j ($j = 1, 2, \dots, k$) are called the *partial regression coefficients*.

The parameter β_j represents the expected change in the response y per unit change in x_j when all the remaining regressor variables $x_i (i \neq j)$ are held constant. For this reason the parameters $\beta_j (j= 1, 2, \dots, k)$ are called the partial regression coefficients.

Introduction to Probability:

Probability is a measure of certainty. It is a real number from between 0 and 1. Probability of an impossible event is zero and the probability of a sure event is one.

Sure Event:

Experiment:

Tossing a fair coin.

Event A: Getting either Heads or Tails.

In this case, event A encompasses all possible outcomes (Heads or Tails), and since the coin must land on one of these sides, the probability of event A is 1.

$$P(A) = 1.$$

Impossible Event:

Experiment:

Rolling a six-sided die and getting a number greater than 6.

Event B: Getting a number greater than 6.

Since the die has only six sides numbered from 1 to 6, it is impossible to roll a number greater than 6. Therefore, the probability of event B is 0.

$$P(B) = 0.$$

In summary, for a sure event, the probability is 1 because it is certain to occur, while for an impossible event, the probability is 0 because there is no chance of it occurring.

Definition of probability:

Classical or Mathematical definition:

The classical approach applies when each outcome in the sample space is equally likely. Probability is calculated as the ratio of the number of favourable outcomes to the total number of possible outcomes.

Example: Tossing a fair coin.

$$P(\text{Heads}) = 0.5$$

Experiment:

An experiment is an action, process, or phenomenon whose outcome is uncertain.

Deterministic Experiment: Every time the experiment repeated the same result.

Example: Counting the number of stairs of a particular building.

Random Experiment: More than one possible outcome.

Example: Coin tossing, Die throwing, Counting the number of calls that came to a mobile in a day.

Sample Space:

The sample space, often denoted by S , is the set of all possible outcomes of a random experiment. It represents the full range of possible results that could occur.

Example: Rolling a six-sided die.

Sample Space S : {1, 2, 3, 4, 5, 6}

In this case, the sample space includes all the possible outcomes when rolling the die, which are the numbers 1 through 6.

Event:

An event is a subset of the sample space, representing a particular outcome or a collection of outcomes of interest. Events are the specific occurrences that we are interested in or want to analyse.

Example: Tossing a fair coin.

Sample Space S : {Heads, Tails}

Event A : Getting a Heads, denoted as $A = \{\text{Heads}\}$

Here, the event A represents the outcome of getting Heads when tossing the coin. It is a subset of the sample space.

Mutually Exclusive Event:

Two events are mutually exclusive if they cannot occur at the same time; in other words, they have no outcomes in common.

Example: Rolling a six-sided die.

Sample Space S : {1, 2, 3, 4, 5, 6}

Event C : Getting an even number, denoted as $C = \{2, 4, 6\}$

Event D : Getting an odd number, denoted as $D = \{1, 3, 5\}$

Events C and D are mutually exclusive because no outcome (number) can be both even and odd simultaneously.

Rules of addition and multiplication

Rules of Addition:

The addition rule states the probability of two events is the sum of the probabilities of two events that will happen minus the probability of both the events that will happen. Mathematically, the addition rule of probability is expressed as:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

When A and B are mutually exclusive;

$$P(A \cup B) = P(A) + P(B)$$

Experiment: Drawing a single card from a standard deck of 52 cards.

Non-Mutually Exclusive Events:

Events:

C: Drawing a face card (king, queen, or jack).

D: Drawing a red card.

Solution:

$$P(C \cup D) = P(C) + P(D) - P(C \cap D)$$

$$= \frac{12}{52} + \frac{26}{52} - \frac{6}{52}$$

$$= \frac{8}{13}$$

In this case, drawing a face card and drawing a red card are not mutually exclusive because a single card can be both a face card and red (e.g., a red queen). The subtraction term $P(C \cap D)$ accounts for the double counting of red face cards.

Mutually Exclusive Events:

Events:

A: Drawing a heart.

B: Drawing a spade.

Solution:

$$P(A \cup B) = P(A) + P(B)$$

$$= \frac{13}{52} + \frac{13}{52}$$

$$= \frac{1}{2}$$

In this case, drawing a heart and drawing a spade are mutually exclusive because a single card cannot be both a heart and a spade. The sum of their probabilities equals 0.5 .

Rules of Multiplication:

According to the multiplication rule of probability, the probability of occurrence of both the events A and B is equal to the product of the probability of B occurring and the conditional probability that event A occurs given that event B occurs.

If A and B are *dependent events*, then the probability of both events occurring simultaneously is given by:

$$P(A \cap B) = P(B) \cdot P(A|B)$$

$$\text{Or, } P(A \cap B) = P(A) \cdot P(B|A)$$

If A and B are two *independent events* in an experiment, then the probability of both events occurring simultaneously is given by:

$$P(A \cap B) = P(A) \cdot P(B)$$

Drawing two balls successively from a bag containing red and white balls without replacement.

The bag contains 4 red balls and 3 white balls.

Question:

What is the probability of drawing a red ball on the first draw and a white ball on the second draw?

Solution:

Let's denote the events as follows:

R_1 : Drawing a red ball on the first draw.

W_2 : Drawing a white ball on the second draw.

The probability of drawing a red ball on the first draw $P(R_1)$ is the ratio of the number of red balls to the total number of balls before the first draw:

$$P(R_1) = 4/7$$

Now, after drawing one ball, there are 6 balls left in the bag (3 red and 3 white). The probability of drawing a white ball on the second draw given that a red ball was drawn on the first draw

$$P(W_2|R_1) = 3/6$$

The probability of both events happening (drawing a red ball on the first draw and a white ball on the second draw) is the product of their individual probabilities:

$$\begin{aligned} P(R_1 \cap W_2) &= P(R_1) \cdot P(W_2|R_1) \\ &= \frac{4}{7} \times \frac{3}{6} \end{aligned}$$

Now, if drawing two balls successively from a bag containing red and white balls with replacement. As the ball is replaced after the first draw, the probability of drawing a white ball on the second draw $P(W_2)$ is also the ratio of the number of white balls to the total number of balls:

$$\text{So, } P(W_2) = 3/7$$

The probability of both independent events happening (drawing a red ball on the first draw and a white ball on the second draw) is the product of their individual probabilities:

$$\begin{aligned} P(R_1 \cap W_2) &= P(R_1) \cdot P(W_2) \\ &= \frac{4}{7} \times \frac{3}{7} \end{aligned}$$

Conditional probability:

Conditional probability is the probability of an event occurring given that another event has already occurred. It is denoted by $P(A|B)$, read as "the probability of event A given event B."

Formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(A \cap B)$: Probability of both events A and B occurring.

$P(B)$: Probability of event B occurring.

$$\text{Similarly, } P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Example-1:

In a school 30% of the students are female. 10% of female students are colour blind. If a student is selected at random. What is the probability that the student will be colour blind and female?

Solution:

$$\text{Here, } P(F) = 0.30$$

F for female

$$P(C|F) = 0.10$$

C for colour blind

$$P(C \cap F) = ?$$

$$\begin{aligned}\text{We know, } P(C|F) &= \frac{P(C \cap F)}{P(F)} \\ \therefore P(C \cap F) &= P(C|F) \cdot P(F) \\ &= 0.10 \times 0.30 \\ &= 0.03\end{aligned}$$

Example-2:

In a school 30% of the students are female. 3% of students are colour blind females. If a student selected randomly is female. What is the probability that she is colour blind?

Solution:

$$P(F) = 0.30$$

F for female

$$P(C \cap F) = 0.03$$

C for colour blind

$$P(C|F) = ?$$

$$\begin{aligned}\text{We know, } P(C|F) &= \frac{P(C \cap F)}{P(F)} \\ &= \frac{0.03}{0.30} \\ &= 0.10\end{aligned}$$

Independence of events:

Two events A and B are independent if occurrence of B does not change probability A. That means,

$$P(A|B) = P(A)$$

Example-1:

20% of families have cars, 10% families have washing machines and 7% families have both. Does having a washing machine depend on having a car?

Solution:

$$P(C) = 0.20$$

W for washing machine

$$P(W) = 0.10$$

C for cars

$$P(C \cap W) = 0.07$$

$$\begin{aligned}P(W|C) &= \frac{P(C \cap W)}{P(C)} \\ &= \frac{0.07}{0.20} \\ &= 0.35 \neq P(W)\end{aligned}$$

So, the two events are not independent. That means, having washing machines depends on having a car.

Another method:

$$P(C) \cdot P(W) = 0.20 \times 0.10 = 0.02 \neq P(C \cap W)$$

Bayes' Theorem:

Bayes' Theorem is a fundamental concept in probability theory named after Reverend Thomas Bayes. It provides a systematic way to update the probability of a hypothesis in light of new evidence.

If A and B are two events, then the formula for the Bayes theorem is given by:

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A) \cdot P(B|A)}{P(B)} \end{aligned}$$

Priori Probability: The probability $P(A)$ is considered as the priori probability.

Posteriori Probability: The probability $P(A|B)$ is considered as the posteriori probability.

Example-1:

A blood test gives 97% positive (+) results for diseased people and 2% positive results for healthy people. In a community 1% of people have diseases. A person is randomly selected from the community and gives the test.

i. What is the probability that the test will be positive?

ii. If the test is positive, what is the probability that the person has the disease?

Solution:

+ for positive test result

$$P(+|D) = 0.97$$

D for disease

$$P(+|H) = 0.02$$

H for healthy

$$P(D) = 0.01$$

$$P(H) = 0.99$$

$$\begin{aligned} \text{i. } P(+) &= P(D \cap +) + P(H \cap +) \\ &= P(D) \cdot P(+|D) + P(H) \cdot P(+|H) \\ &= 0.01 \times 0.97 + 0.99 \times 0.02 \\ &= 0.0295 \end{aligned}$$

$$\begin{aligned} \text{ii. } P(D|+) &= \frac{P(D \cap +)}{P(+)} \\ &= \frac{P(D) \cdot P(+|D)}{P(+)} \\ &= \frac{0.01 \times 0.97}{0.0295} \\ &= 0.329 \end{aligned}$$

Example-2:

3 Machines produce disks. First machine produces 30%, of which 5% are defective. Second machine produces 50%, of which 3% are defective. Third machine produces 20%, of which 10% are defective. A disk is randomly selected from the mixed lot.

i. What is the probability that the disk is defective?

ii. If the disk is defective, what is the probability that it was produced by the second machine?

Solution:

D for defective disk

$$P(M_1) = 0.30$$

M1 for first machine

$$P(M_2) = 0.50$$

M2 for second machine

$$P(M_3) = 0.20$$

M3 for third machine

$$P(D|M_1) = 0.05$$

$$P(D|M_2) = 0.03$$

$$P(D|M_3) = 0.10$$

$$\begin{aligned} \text{i. } P(D) &= P(D \cap M_1) + P(D \cap M_2) + P(D \cap M_3) \\ &= P(M_1) \times P(D|M_1) + P(M_2) \times P(D|M_2) + P(M_3) \times P(D|M_3) \\ &= 0.30 \times 0.05 + 0.50 \times 0.03 + 0.20 \times 0.10 \\ &= 0.05 \end{aligned}$$

$$\begin{aligned} \text{iii. } P(M_2|D) &= \frac{P(D \cap M_2)}{P(D)} \\ &= \frac{P(M_2) \cdot P(D|M_2)}{P(D)} \\ &= \frac{0.50 \times 0.03}{0.05} \\ &= 0.30 \end{aligned}$$

Example-3:

Bag I contains 4 white and 6 black balls while Bag II contains 4 white and 3 black balls. One ball is drawn at random from one of the bags, and it is found to be black. Find the probability that it was drawn from Bag I.

Solution:

B for choosing black ball

$$P(B_1) = 0.50$$

B₁ for choosing bag I

$$P(B_2) = 0.50$$

B₂ for choosing bag II

$$P(B|B_1) = 6/10 = 0.60$$

$$P(B|B_2) = 3/7 = 0.43$$

$$\begin{aligned} P(B_1|B) &= \frac{P(B_1 \cap B)}{P(B)} \\ &= \frac{P(B_1) \cdot P(B|B_1)}{P(B_1 \cap B) + P(B_2 \cap B)} \\ &= \frac{P(B_1) \cdot P(B|B_1)}{P(B_1) \cdot P(B|B_1) + P(B_2) \cdot P(B|B_2)} \\ &= \frac{0.50 \times 0.60}{0.50 \times 0.60 + 0.50 \times 0.43} \\ &= 0.583 \end{aligned}$$

Random Variable:

A random variable is a function that assigns a numerical value to each outcome of a random experiment.

Example: X = Number of heads out of two tossed of a fair coin.

So, $X = (0, 1, 2)$

Types of Random Variable:

There are two types of random variables:

(a) Discrete random variable:

If a random variable X assumes only a *finite number* of values, then it is called a discrete random variable. Example: Number of throws of a fair coin before the first head occurs.

(b) Continuous random variable:

If a random variable is such that it assumes any value within a given interval, then it is called a continuous random variable.

Example: The heights of the persons collected from a crowd.

Probability mass function (Pmf):

A function $p(x)$ that gives probability for different values of a discrete random variable X .

Properties of $p(x)$:

- i. $p(x) \geq 0$
- ii. $\sum_x p(x) = 1$

Example: Two fair coins tossed once.

X = number of heads out of 2 tossed

So, $X = (0, 1, 2)$

Here, $S = \{HH, HT, TH, TT\}$

$$P(0) = 1/4$$

$$P(1) = P(HT) + P(TH)$$

$$= 1/4 + 1/4$$

$$= 1/2$$

$$P(2) = 1/4$$

x	0	1	2
$p(x)$	1/4	1/2	1/4

Probability density function (Pdf): Let X be a random variable. A function $f(x)$ is called pdf of X if it satisfied

- i. $f(x) \geq 0$
- ii. $\int_{-\infty}^{\infty} f(x) dx = 1$

Example: $f(x) = k(1 - x^2)$; $-1 < x < 1$ then Determine,

- i. k
- ii. $P(-0.5 < x < 0.5)$
- iii. $P(x < 0)$
- iv. $P(x > 0.5)$
- v. $P(x = 0.7)$

Solution:

$$i. \int_{-1}^1 k(1 - x^2) dx = 1$$

$$\Rightarrow k \left(x - \frac{x^3}{3} \right)_{-1}^1 = 1$$

$\vdots$

$\vdots$

$$\therefore k = \frac{3}{4}$$

$$\therefore f(x) = \frac{3}{4}(1 - x^2) \quad -1 < x < 1$$

$$ii. \int_{-0.5}^{0.5} \frac{3}{4}(1 - x^2) dx$$

$$= \frac{3}{4} \left(x - \frac{x^3}{3} \right)_{-0.5}^{0.5}$$

$\vdots$

$\vdots$

$=$

$$\begin{aligned} \text{iii. } & \int_{-1}^0 \frac{3}{4}(1-x^2) dx \\ &= \frac{3}{4} \left(x - \frac{x^3}{3} \right)_{-1}^0 \end{aligned}$$

$$\vdots$$

$$\vdots$$

$$=$$

$$\begin{aligned} \text{iv. } & \int_{.5}^1 \frac{3}{4}(1-x^2) dx \\ &= \frac{3}{4} \left(x - \frac{x^3}{3} \right)_{.5}^1 \end{aligned}$$

$$\vdots$$

$$\vdots$$

$$=$$

$$\text{v. } P(x = 0.7) = 0$$

As, in this case length = .7 but width = 0.

So, area = length $\times$ width = 0 [Integration gives area]

Mathematical Expectation:

Let us consider a discrete random variable X which assumes the values $x_1, x_2, \dots, x_n$ with respective probabilities $p_1, p_2, \dots, p_n$, such that

$$\sum_{i=1}^n p_i = 1$$

, then the mathematical expectation of the random variable X is given by the sum of the products of the different values of X with their corresponding probabilities. The expectation of a random variable is generally denoted by $E(X)$.

$$E(X) = \sum_{i=1}^n x_i p_i$$

If X is a continuous random variable with probability density function

$$f(x) \quad -\infty < x < \infty$$

Then the mathematical expectation of the random variable X is given by

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Some properties of Mathematical Expectations:

- i. If C is a constant then $E(C) = C$
- ii. If C is a constant then $E(CX) = C E(X)$
- iii. If a and c are two constants then $E(aX + c) = a E(X) + c$
- iv. The expectation of the sum of two random variables are equal to the sum of their expectations i.e. $E(X + Y) = E(X) + E(Y)$
- v. The expectation of the difference of two random variables are equal to the difference of their expectations i.e. $E(X - Y) = E(X) - E(Y)$
- vi. If X and Y are two independent random variables then, $E(XY) = E(X) E(Y)$

Variance of a Random Variable:

Variance is an important characteristic of a random variable. It is a measure of dispersion of the random variable. Variance of X, is denoted by $Var(X)$ or by σ_x^2 . It is defined as the expected value of the square of the deviation of the random variable from its mean value.

$$Var(X) = E[X - E(X)]^2$$

The positive square-root of the variance is called the standard deviation and is denoted by σ_x .

A simplified expression for the variance can be derived as:

$$Var(X) = E(X^2) - \{E(X)\}^2$$

Some properties of variance of a Random Variable:

- i. If C is a constant then $Var(C) = 0$
- ii. If C is a constant then we have $Var(CX) = C^2 Var(X)$
- iii. If A and C are two constants then $Var(AX + C) = A^2 Var(X)$
- iv. If X and Y are *two independent* random variables then,

$$Var(X + Y) = Var(X) + Var(Y)$$

$$Var(X - Y) = Var(X) + Var(Y)$$
- v. If X and Y are *dependent* random variables then,

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

$$Var(X - Y) = Var(X) + Var(Y) - 2Cov(X, Y)$$

Example-1:

Let X be a random variable with probability distribution

x	0	1	2	3
p(x)	1/3	1/2	0	1/6

Determine,

- a) $E(X)$
- b) $E(X^2)$
- c) $E[(X - 1)^2]$
- d) $P(X > 1)$
- e) $P(X \leq 2)$
- f) $\text{Var}(X)$

Solution:

$$\begin{aligned} \text{a) } E(X) &= \sum_i^n x_i p_i \\ &= 0 \times \frac{1}{3} + 1 \times \frac{1}{2} + 2 \times 0 + 3 \times \frac{1}{6} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{b) } E(X^2) &= \sum_i^n x_i^2 p_i \\ &= 0^2 \times \frac{1}{3} + 1^2 \times \frac{1}{2} + 2^2 \times 0 + 3^2 \times \frac{1}{6} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{c) } E[(X - 1)^2] &= \sum_i^n (x_i - 1)^2 p_i \\ &= (0 - 1)^2 \times \frac{1}{3} + (1 - 1)^2 \times \frac{1}{2} + (2 - 1)^2 \times 0 + (3 - 1)^2 \times \frac{1}{6} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{d) } P(X > 1) &= P(X = 2) + P(X = 3) \\ &= 0 + \frac{1}{6} \\ &= \frac{1}{6} \end{aligned}$$

$$\begin{aligned} \text{e) } P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= \frac{1}{3} + \frac{1}{2} + 0 \\ &= \frac{5}{6} \end{aligned}$$

$$\begin{aligned} \text{f) } \text{Var}(X) &= E(X^2) - \{E(X)\}^2 \\ &= 2 - 1^2 \\ &= 1 \end{aligned}$$

Example-2:

$$\begin{aligned} f(x) &= 2x, 0 \leq x \leq 1 \\ E(X) &=? \\ \text{V}(X) &=? \end{aligned}$$

Solution:

$$E(X) = \int_0^1 x f(x) dx$$

$$= \int_0^1 x \cdot 2x dx$$

$$= 2 \int_0^1 x^2 dx$$

$$= 2 \left[\frac{x^3}{3} \right]_0^1$$

$$= 2 \left[\frac{1}{3} - 0 \right]$$

$$= \frac{2}{3}$$

$$E(X^2) = \int_0^1 x^2 f(x) dx$$

Then,

$$Var(X) = E(X^2) - \{E(X)\}^2$$

Example-3:

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 2 - x, & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = ?$$

$$V(X) = ?$$

Solution:

$$E(X) = \int_0^2 x f(x) dx$$

$$= \int_0^1 x \cdot x dx + \int_1^2 x \cdot (2 - x) dx$$

$$= \int_0^1 x^2 dx + \int_1^2 (2x - x^2) dx$$

∴

∴

$$= \frac{1}{3} + \frac{2}{3}$$

$$= 1$$

$$\begin{aligned}
 E(X^2) &= \int_0^2 x^2 f(x) dx \\
 &= \int_0^1 x^2 \cdot x dx + \int_1^2 x^2 \cdot (2-x) dx \\
 &= \int_0^1 x^3 dx + \int_1^2 (2x^2 - x^3) dx
 \end{aligned}$$

Then,

$$Var(X) = E(X^2) - \{E(X)\}^2$$

Joint Distribution:

In probability theory and statistics, the joint distribution of two or more random variables describes how the probabilities of these variables are related.

Joint probability mass function (joint pmf):

Let's consider two discrete random variables, X and Y. The joint probability mass function (joint pmf), denoted as $p(x, y)$, gives the probability of both $X = x$ and $Y = y$ occurring simultaneously.

Properties of $p(x, y)$

- i. $P(x, y) \geq 0$
- ii. $\sum_x \sum_y P(x, y) = 1$

Example-1

$x \backslash y$	0	1	2	3	Total
0	4/23	2/23	1/23	1/23	8/23
1	2/23	1/23	2/23	1/23	6/23
2	0	1/23	7/23	1/23	9/23
Total	6/23	4/23	10/23	3/23	

$$P(0, 0) = 4/23$$

$$P(1, 0) = 2/23$$

Marginal pmf:

$$p_x(x) = \sum_y p(x, y)$$

$$p_y(y) = \sum_x p(x, y)$$

$$\text{Here, } p_x(0) = \frac{4}{23} + \frac{2}{23} + \frac{1}{23} + \frac{1}{23} = \frac{8}{23}$$

$$p_x(1) = \frac{2}{23} + \frac{1}{23} + \frac{2}{23} + \frac{1}{23} = \frac{6}{23}$$

$$p_x(2) = 0 + \frac{1}{23} + \frac{7}{23} + \frac{1}{23} = \frac{9}{23}$$

∴ Marginal pmf of X is

x	0	1	2
$p_x(x)$	8/23	6/23	9/23

$$\text{Now, } p_y(0) = \frac{4}{23} + \frac{2}{23} + 0 = \frac{6}{23}$$

$$p_y(1) = \frac{2}{23} + \frac{1}{23} + \frac{1}{23} = \frac{4}{23}$$

$$p_y(2) = \frac{1}{23} + \frac{2}{23} + \frac{7}{23} = \frac{10}{23}$$

$$p_y(3) = \frac{1}{23} + \frac{1}{23} + \frac{1}{23} = \frac{3}{23}$$

∴ Marginal pmf of Y is

y	0	1	2	3
$p_y(y)$	6/23	4/23	10/23	3/23

Example-2

$x \backslash y$	1	2	5	7	Total
0	1/13	3/13	1/13	2/13	7/13
1	1/13	1/13	k	2/13	(4+13k)/13
Total	2/13	4/13	(1+13k)/13	4/13	

Determine

- Find k
- Marginal pmf of Y
- $p(y = 5)$

Solution:

$$\text{i. } \sum p_x(x) = \frac{7}{13} + \frac{4+13k}{13} = 1$$

$$\therefore k = \frac{2}{13}$$

Similarly,

$$\sum p_y(y) = \frac{2}{13} + \frac{4}{13} + \frac{1+13k}{13} + \frac{4}{13} = 1$$

$$\therefore k = \frac{2}{13}$$

ii. Using $k = \frac{2}{13}$ in the joint pmf table

$x \backslash y$	1	2	5	7	Total
0	1/13	3/13	1/13	2/13	7/13
1	1/13	1/13	2/13	2/13	6/13
Total	2/13	4/13	3/13	4/13	

$\therefore$ Marginal pmf of Y is

y	1	2	5	7
$p_y(y)$	2/13	4/13	3/13	4/13

$$\text{iii. } p(y = 5) = p_y(5) = \frac{1}{13} + \frac{2}{13} = \frac{3}{13}$$

Conditional pmf:

$$p_{x|y}(x|y) = \frac{p(x, y)}{p_y(y)}$$

Example

$x \backslash y$	0	1	2	Total
1	1/17	3/17	3/17	7/17
2	3/17	5/17	2/17	10/17
Total	4/17	8/17	5/17	

Determine

- i. $p(x = 1)$
- ii. $p(y = 2)$
- iii. $p_{x|y}(1|2)$
- iv. Conditional pmf of y given x equal to 1

Solution:

$$\text{i. } p(x = 1) = p_x(1) = \frac{1}{17} + \frac{3}{17} + \frac{3}{17} = \frac{7}{17}$$

$$\text{ii. } p(y = 2) = p_y(2) = \frac{3}{17} + \frac{2}{17} = \frac{5}{17}$$

$$\begin{aligned} \text{iii. } p_{x|y}(1|2) &= \frac{p(1,2)}{p_y(2)} \\ &= \frac{\frac{3}{17}}{\frac{5}{17}} = \frac{3}{5} \end{aligned}$$

$$\begin{aligned} \text{iv. } p_{y|x}(0|1) &= \frac{p(1,0)}{p_x(1)} \\ &= \frac{\frac{1}{17}}{\frac{7}{17}} = \frac{1}{7} \end{aligned} \quad \left[p_x(1) = \frac{7}{17} \text{ from i} \right]$$

$$\begin{aligned} p_{y|x}(1|1) &= \frac{p(1,1)}{p_x(1)} \\ &= \frac{\frac{3}{17}}{\frac{7}{17}} = \frac{3}{7} \end{aligned}$$

$$\begin{aligned} p_{y|x}(2|1) &= \frac{p(1,2)}{p_x(1)} \\ &= \frac{\frac{3}{17}}{\frac{7}{17}} = \frac{3}{7} \end{aligned}$$

$\therefore$ The conditional pmf of y given x equal to 1

y	0	1	2
$p_{y x}(y 1)$	1/7	3/7	3/7

Joint probability distribution function (joint pdf):

Let's consider two continuous random variables, X and Y .

$f(x, y)$ is called joint pdf of X and Y if

- i. $f(x, y) \geq 0$
- ii. $\int_x \int_y f(x, y) dy dx = 1$

Example-1

$$f(x, y) = 6e^{-2x}e^{-3y} \quad x > 0, y > 0$$

- i. Check whether $f(x, y)$ is a joint pdf.
- ii. Calculate $p(x < 1, y < 2)$

Solution:

$$\begin{aligned}
 \text{i.} \quad & \int_0^{\infty} \int_0^{\infty} 6e^{-2x}e^{-3y} dy dx \\
 &= \int_0^{\infty} 6e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_0^{\infty} dx \\
 &= \int_0^{\infty} -2e^{-2x}[0 - 1] dx \\
 &= \int_0^{\infty} 2e^{-2x} dx \\
 &= 2 \left[\frac{e^{-2x}}{-2} \right]_0^{\infty} \\
 &= -[0 - 1] \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \text{ii.} \quad & \int_0^1 \int_0^2 6e^{-2x}e^{-3y} dy dx \\
 &= \int_0^1 6e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_0^2 dx \\
 &= \int_0^1 -2e^{-2x}[e^{-6} - 1] dx \\
 &= -2[e^{-6} - 1] \int_0^1 e^{-2x} dx \\
 &= -2[e^{-6} - 1] \left[\frac{e^{-2x}}{-2} \right]_0^1 \\
 &= [e^{-6} - 1][e^{-2} - 1] \\
 &= 0.86
 \end{aligned}$$

Marginal pdf:

Marginal pdf of X

$$f_x(x) = \int_y f(x, y) dy$$

Marginal pdf of Y

$$f_y(y) = \int_x f(x, y) dx$$

$$\begin{aligned} \text{iii. } f_x(x) &= \int_0^{\infty} 6e^{-2x} e^{-3y} dy \\ &= 6e^{-2x} \left[\frac{e^{-3y}}{-3} \right]_0^{\infty} \\ &= -2e^{-2x} [0 - 1] \\ &= 2e^{-2x} \\ \therefore f_x(x) &= 2e^{-2x} \quad x > 0 \end{aligned}$$

$$\begin{aligned} \therefore p(x < 3) &= \int_0^3 f_x(x) dx \\ &= \int_0^3 2e^{-2x} dx \\ &= 2 \left[\frac{e^{-2y}}{-2} \right]_0^3 \\ &= -[e^{-6} - 1] \\ &= [1 - e^{-6}] \\ &= 0.99 \end{aligned}$$

Example-2

$$f(x, y) = x + y \quad 0 < x < 1, \quad 0 < y < 1$$

- i. Find the marginal pdf of X and Y
- ii. Calculate $p(.3 < x < .7)$
- iii. Calculate $p(.5 < y < .9)$

Solution:

$$\begin{aligned} \text{i. } f_x(x) &= \int_0^1 (x + y) dy \\ &= \int_0^1 x dy + \int_0^1 y dy \end{aligned}$$

$$\begin{aligned}
 &= x[y]_0^1 + \left(\frac{y^2}{2}\right)_0^1 \\
 &= x + \frac{1}{2} \\
 \therefore f_x(x) &= x + \frac{1}{2} \quad 0 < x < 1
 \end{aligned}$$

$$\begin{aligned}
 f_y(y) &= \int_0^1 (x + y) dx \\
 &= \int_0^1 x dx + \int_0^1 y dx \\
 &= \left(\frac{x^2}{2}\right)_0^1 + y[x]_0^1 \\
 &= y + \frac{1}{2} \\
 \therefore f_y(y) &= y + \frac{1}{2} \quad 0 < y < 1
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } p(.3 < x < .7) &= \int_{.3}^{.7} f_x(x) dx \\
 &= \int_{.3}^{.7} \left(x + \frac{1}{2}\right) dx
 \end{aligned}$$

$$\begin{aligned}
 \text{iii. } p(.5 < y < .9) &= \int_{.5}^{.9} f_y(y) dy \\
 &= \int_{.5}^{.9} \left(y + \frac{1}{2}\right) dy
 \end{aligned}$$

Conditional pdf:

$$\begin{aligned}
 f_{y|x}(y|x) &= \frac{f(x, y)}{f_x(x)} \\
 f_{x|y}(x|y) &= \frac{f(x, y)}{f_y(y)}
 \end{aligned}$$

From example 2

- i. $f_{y|x}(y|0.7)$
- ii. $p(.3 < y < .6|x = 0.7)$
- iii. $p(.2 < x < .4|y = 0.5)$

Solution:

$$\text{i. } f_{y|x}(y|0.7) = \frac{f(0.7, y)}{f_x(0.7)}$$

$$\begin{aligned}
 f(x, y) &= x + y \quad 0 < x < 1, \quad 0 < y < 1 \\
 \therefore f(0.7, y) &= 0.7 + y \\
 f_x(x) &= x + \frac{1}{2} \quad 0 < x < 1 \\
 \therefore f_x(0.7) &= 0.7 + \frac{1}{2} \\
 &= 1.2
 \end{aligned}$$

$$\begin{aligned}\therefore f_{y|x}(y|0.7) &= \frac{0.7 + y}{1.2} \\ &= \frac{1}{1.2}(0.7 + y) \quad 0 < y < 1\end{aligned}$$

$$\text{ii. } p(.3 < y < .6|x = 0.7) = \int_{.3}^{.6} \frac{1}{1.2}(0.7 + y)dy$$

iii. HW

(I will put this HW in assignment 3)

Bernoulli Trials:

A Bernoulli trial is a random experiment with only two possible outcomes: success and failure. The key characteristics of a Bernoulli trial are:

1. Two Outcomes: The experiment results in one of two possible outcomes: success and failure.
2. Constant Probability: The probability of success, denoted as p , remains constant for each trial.
3. Independent Trials: The outcome of one trial does not affect the outcome of another.

Binomial Distribution:

The Binomial distribution arises when we conduct a fixed number n of independent Bernoulli trials, each with the same probability of success p . The random variable X represents the number of successes in these trials. Then

Then $X \sim \text{binomial}(n, p)$ [X follows binomial distribution]

Here, n and p are two parameters of binomial distribution.

The probability mass function (pmf) of the Binomial distribution is

given by: $p(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, 2, \dots, n$

Where:

- ✓ n is the number of trials.
- ✓ x is the number of successes.
- ✓ p is the probability of success in a single trial.
- ✓ $(1 - p)$ is the probability of failure in a single trial.
- ✓ $\binom{n}{x}$ is the binomial coefficient.

Mean and Variance:

$$E(X) = np$$

$$V(X) = np(1 - p)$$

Problem:

An MCQ test has 10 questions each with 5 possible choices. A student did not study at all and chose all the answers randomly.

- i. Probability that 2 answers will be correct.
- ii. Probability that at least 1 answer will be correct.
- iii. Probability that at most 2 answers will be correct.
- iv. Probability that less than 1 answer will be correct.
- v. Average number of correct answers.
- vi. Variance of this distribution.

Solution:

Here, X = number of successes

$$\therefore X \sim \text{binomial } (n = 10, p = \frac{1}{5} = 0.2)$$

$$\therefore p(X = x) = \binom{10}{x} 0.2^x (1 - 0.2)^{10-x} \quad x = 0, 1, 2, \dots, 10$$

$$\text{i. } p(X = 2) = \binom{10}{2} 0.2^2 (1 - 0.2)^{10-2}$$

$$\begin{aligned} \text{ii. } p(X \geq 1) &= 1 - p(X = 0) \\ &= 1 - \binom{10}{0} 0.2^0 (1 - 0.2)^{10-0} \end{aligned}$$

$$\text{iii. } p(X \leq 2) = p(X = 0) + p(X = 1) + p(X = 2)$$

$$\text{iv. } p(X < 1) = p(X = 0)$$

$$\text{v. } E(X) = np = 10 * 0.2 = 2$$

$$\text{vi. } V(X) = np(1 - p) = 10 * 0.2 * (1 - 0.2) = 1.6$$

Practice problems:

(a) A factory produces light bulbs, and 90% of them pass quality control. If a sample of 8 bulbs is randomly selected, what is the probability that exactly 7 pass quality control?

Solution:

X = number of successes.

$\therefore X \sim \text{binomial } (n = 8, p = 0.90)$

$$\therefore p(X = x) = \binom{8}{x} 0.9^x (1 - 0.9)^{8-x} \quad x = 0, 1, 2, \dots, 8$$

$$\text{So, } p(X = 7) = \binom{8}{7} 0.9^7 (1 - 0.9)^{8-7}$$

(b) A basketball player has a free throw success rate of 80%. If the player takes 7 free throws, what is the probability of making at least 5 of them?

Solution:

X = number of successes

$\therefore X \sim \text{binomial } (n = 7, p = 0.80)$

$$\therefore p(X = x) = \binom{7}{x} 0.8^x (1 - 0.8)^{7-x} \quad x = 0, 1, 2, \dots, 7$$

$$\text{So, } p(X \geq 5) = p(X = 5) + p(X = 6) + p(X = 7)$$

Geometric Distribution:

The Geometric distribution models the number of Bernoulli trials needed for the first success. It is derived from a sequence of independent Bernoulli trials, each with the same probability of success p . The random variable X represents the number of trials required until the first success.

Then,

$X \sim \text{geometric } (p)$

[X follows geometric distribution]

Here, p is the parameter of geometric distribution.

The probability mass function (pmf) of the Geometric distribution is given by: $p(X = x) = p(1 - p)^{x-1}$ $x = 1, 2, 3, \dots$

Where:

- ✓ x is the number of trials required until the first success.
- ✓ p is the probability of success in a single trial.

Mean and Variance:

$$E(X) = \frac{1}{p}$$

$$V(X) = \frac{1-p}{p^2}$$

Problem:

A die is thrown until 6 occurs for the first time. What is the probability that

- i. Exactly 3 tosses will be required.
- ii. More than two tosses will be required.
- iii. Find the average number of tosses required.
- iv. Find the variance of this distribution.

Solution:

X = number of trials required until 6 occurs for the first time.

$\therefore X \sim \text{geometric} (p = \frac{1}{6})$

$$\therefore p(X = x) = \frac{1}{6} \left(1 - \frac{1}{6}\right)^{x-1} \quad x = 1, 2, 3, \dots$$

$$\begin{aligned} \text{i. } p(X = 3) &= \frac{1}{6} \left(1 - \frac{1}{6}\right)^{3-1} \\ &= \frac{1}{6} \left(\frac{5}{6}\right)^2 \\ &= \frac{25}{36} \end{aligned}$$

$$\text{ii. } p(X > 2) = 1 - p(X = 1) - p(X = 2)$$

$$\text{iii. } E(X) = \frac{1}{p} = \frac{1}{\frac{1}{6}} = 6$$

$$\text{iv. } V(X) = \frac{1-p}{p^2} = \frac{1-\frac{1}{6}}{\left(\frac{1}{6}\right)^2}$$

Practice problems:

(a) A biased coin has a probability of landing heads equal to 0.2. What is the probability that the first head occurs on the 4th toss?

Solution:

X = number of trials required until head occurs for the first time.

$\therefore X \sim \text{geometric } (p=0.2)$

$$\therefore p(X = x) = 0.2(1 - 0.2)^{x-1} \quad x = 1, 2, 3, \dots$$

$$\text{So, } p(X = 4) = 0.2(1 - 0.2)^{4-1}$$

(b) A student is preparing for a multiple-choice exam. The student knows that the probability of guessing the correct answer to a question is 0.25. What is the probability that the student guesses the correct answer on the 5th attempt?

Solution:

X = number of attempts required until guessing the correct answer to a question.

$\therefore X \sim \text{geometric } (p=0.25)$

$$\therefore p(X = x) = 0.25(1 - 0.25)^{x-1} \quad x = 1, 2, 3, \dots$$

$$\text{So, } p(X = 5) = 0.25(1 - 0.25)^{5-1}$$

Poisson distribution:

A probability distribution that models the number of events that occur in a fixed interval of time or space under the assumption that these events occur with a known constant mean rate and are independent of the time since the last event. Usually rare events.

The random variable X represents the number of events in an interval of time or in a region.

Then $X \sim \text{poisson } (\lambda)$ [X follows poisson distribution]

Here, rate of event λ is the parameter of geometric distribution.

The probability mass function (pmf) of the poisson distribution is given

$$\text{by: } p(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots$$

Where:

- ✓ x is the number of events in an interval of time or in a region.
- ✓ λ is the rate of events.

Mean and Variance:

$$E(X) = \lambda$$

$$V(X) = \lambda$$

Problem-1

A call centre receives an average of 5 customer service calls per hour. What is the probability that the call centre receives exactly 3 calls in the next hour?

Solution:

X = number of calls received by a call centre in an hour.

$$\therefore X \sim \text{poisson} (\lambda = 5)$$

$$\therefore p(X = x) = \frac{e^{-5} 5^x}{x!} \quad x = 0, 1, 2, \dots \dots \dots$$

$$\text{So, } p(X = 3) = \frac{e^{-5} 5^3}{3!} = 0.1404$$

Problem-2:

On average, a factory experiences 2 machine failures per day. What is the probability that the factory will have no machine failures on a given day?

Solution:

X = number of machine failures on a given day.

$$\therefore X \sim \text{poisson} (\lambda = 2)$$

$$\therefore p(X = x) = \frac{e^{-2} 2^x}{x!} \quad x = 0, 1, 2, \dots \dots \dots$$

$$\text{So, } p(X = 0) = \frac{e^{-2} 2^0}{0!} = 0.1353$$

Problem-3:

A website receives an average of 8 new user registrations per hour. What is the probability that the website receives at least 1 new registrations in the next hour?

Solution:

X = number of new user registrations per hour.

$$\therefore X \sim \text{poisson} (\lambda = 8)$$

$$\therefore p(X = x) = \frac{e^{-8} 8^x}{x!} \quad x = 0, 1, 2, \dots \dots \dots$$

$$\text{So, } p(X \geq 1) = 1 - p(X = 0) = 1 - \frac{e^{-8} 8^0}{0!} = 0.999$$

Exponential Distribution:

The exponential distribution is a continuous probability distribution that describes the time between independent events occurring at a constant rate. It is often used to model the time until the next event in a sequence of events, where events happen continuously and independently at a constant average rate. It is useful for modelling failure times and waiting times.

A random variable X distributed as an exponential random variable with parameter $(\lambda > 0)$, which is written $X \sim \text{exp}(\lambda)$. The pdf of the exponential distribution is given by the formula:

$$f(x) = \lambda e^{-\lambda x}; x \geq 0$$

Where:

- ✓ x is the random variable representing the time between events
- ✓ λ is the rate parameter

Mean and Variance:

$$E(X) = \frac{1}{\lambda} \text{ and } V(X) = \frac{1}{\lambda^2} \quad [\text{HW}]$$

Example-1:

Suppose that you are waiting for a friend to call you and that the time you wait in minutes has an exponential distribution with parameter $\lambda = 0.1$.

- (a) What is the expectation of your waiting time?
 - (b) What is the probability that you will wait longer than 10 minutes?
 - (c) What is the probability that you will wait less than 5 minutes?
-

Solution:

X =time required

$$\therefore X \sim \text{exp} (\lambda = 0.1)$$

$$\therefore f(x) = 0.1e^{-0.1x}; x \geq 0$$

$$(a) \quad E(X) = 1 / 0.1 = 10$$

$$(b) \quad P(X > 10) = \int_{10}^{\infty} f(x)dx = \int_{10}^{\infty} 0.1e^{-0.1x}dx$$

$$= 0.1 \left[\frac{e^{-0.1x}}{-0.1} \right]_{10}^{\infty} = -[0 - e^{-0.1 \cdot 10}] = 0.3679$$

$$(c) \quad P(X < 5) = \int_0^5 f(x)dx = \int_0^5 0.1e^{-0.1x}dx = 0.3935$$

Example-2:

Average time required to repair a machine is 0.5 hours. What is the probability that the next time required will be more than 1 hour?

Solution:

X =time required

$$\therefore X \sim \text{exp} (\lambda = \frac{1}{0.5} = 2)$$

$$\therefore f(x) = 2e^{-2x}; x \geq 0$$

$$\therefore P(X > 1) = \int_1^{\infty} f(x)dx = \int_1^{\infty} 2e^{-2x}dx = 0.1353$$

Example-3:

Radio of a particular brand follows exponential distribution with an average lifetime of 2 years. What is the probability that a randomly chosen radio will survive at least 4 years?

Solution:

X = Lifetime of the radio

$$\therefore X \sim \exp \left(\lambda = \frac{1}{2} = 0.5 \right)$$

$$\therefore f(x) = 0.5e^{-0.5x}; x \geq 0$$

$$\therefore P(X > 4) = \int_4^{\infty} f(x)dx = \int_4^{\infty} 0.5e^{-0.5x}dx = 0.1353$$

Normal Distribution: A normal distribution is a continuous probability distribution that is symmetric around its mean. It is fully characterized by two parameters: the mean (μ) and the variance (σ^2). A random variable X distributed as a normal random variable with parameter μ ($-\infty < \mu < \infty$) and σ^2 ($\sigma^2 > 0$), which is written $X \sim N(\mu, \sigma^2)$. The pdf of the normal distribution is given by the formula:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty$$

Where:

- ✓ μ is the mean and σ^2 is the variance
- ✓ e is the base of the natural logarithm and $\pi = 3.1416$

Mean and Variance:

$$E(X) = \mu \text{ and } V(X) = \sigma^2$$

Standard Normal Distribution: The standard normal distribution is a specific case of the normal distribution where the mean (μ) is 0 and the variance (σ^2) is 1. A random variable following a standard normal distribution is denoted as Z . The probability density function (pdf) for the standard normal distribution is given by the formula:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}; -\infty < z < \infty$$

Values from the standard normal distribution are commonly found in statistical tables or computed using statistical software.

The relationship between a random variable X following a normal distribution with mean μ and standard deviation σ and a corresponding standard normal variable Z is given by the formula: $Z = \frac{X - \mu}{\sigma}$.

Example 1: Suppose that $Z \sim N(0, 1)$. Find: (a) $P(Z < 1.34)$ (b) $P(Z > -0.22)$ (c) $P(-2.19 < Z < 0.43)$ (d) $P(0.09 < Z < 1.76)$ (e) $P(|Z| < 0.38)$ (f) $P(|Z| > 0.38)$

Solution:

$$(a) P(Z < 1.34) = \Phi(1.34) = 0.9099$$

[where $\Phi(Z)$ is the cdf(cumulative distribution function) of Z , $\Phi(Z) = P(Z < z)$]

$$\begin{aligned} (b) \quad P(Z > -0.22) &= 1 - P(Z < -0.22) \\ &= 1 - \Phi(-0.22) = 1 - 0.41294 = 0.58706 \end{aligned}$$

$$\begin{aligned} (c) P(-2.19 < Z < 0.43) &= \Phi(0.43) - \Phi(-2.19) \\ &= 0.6664 - 0.0143 = 0.6521 \end{aligned}$$

$$\begin{aligned} (d) \quad P(0.09 < Z < 1.76) &= \Phi(1.76) - \Phi(0.09) \\ &= 0.9608 - 0.53586 = 0.42494 \end{aligned}$$

$$\begin{aligned} (e) P(|Z| < 0.38) &= P(-0.38 < Z < 0.38) \\ &= \Phi(0.38) - \Phi(-0.38) = 0.6480 - 0.352 \\ &= 0.296 \end{aligned}$$

$$\begin{aligned} (f) P(|Z| > 0.38) &= 1 - P(|Z| < 0.38) \\ &= 1 - 0.296 = 0.704 \end{aligned}$$

Example2: If X is a normal random variable with parameters $\mu = 3$ and $\sigma^2 = 9$, find (a) $P\{2 < X < 5\}$; (b) $P\{X > 0\}$; (c) $P\{|X - 3| > 6\}$.

Solution:

$$(a) \quad P\{2 < X < 5\} = P\left\{\frac{2-3}{3} < \frac{X-3}{3} < \frac{5-3}{3}\right\} = P\left\{\frac{-1}{3} < Z < \frac{2}{3}\right\} = \Phi\left(\frac{2}{3}\right) - \Phi\left(\frac{-1}{3}\right)$$

$$= 0.74751 - 0.36944 = 0.378$$

$$(b) \quad P\{X > 0\} = P\left\{\frac{X-3}{3} > \frac{0-3}{3}\right\} = P\{Z > -1\} = 1 - \Phi(-1) = 1 - 0.15866 = 0.84135$$

$$\begin{aligned} (c) \quad P\{|X - 3| > 6\} &= 1 - P\{|X - 3| < 6\} \\ &= 1 - P\{-6 < X - 3 < 6\} \\ &= 1 - P\left\{\frac{-6}{3} < \frac{X-3}{3} < \frac{6}{3}\right\} \\ &= 1 - P\{-2 < Z < 2\} \\ &= 1 - [\Phi(2) - \Phi(-2)] \\ &= 1 - [0.9772 - 0.0228] \\ &= 0.0456 \end{aligned}$$

The End!!!!
